

Lexicon against Naturalness: Unnatural Gradient Phonotactic Restrictions in Tarma Quechua

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Abstract

It has been shown in separate studies that phonological grammar can operate probabilistically and in phonetically unnatural directions. This paper documents a case in which both of these properties of phonological grammar are true at the same time. We create a new corpus of Tarma Quechua vocabulary (based both on published and unpublished data), and argue that the unnatural probabilistic phonotactic trends in the Tarma Quechua lexicon are statistically significant and show some signs of productivity based on morphophonological alternations. We also perform an acoustic analysis of existing recordings to support the phonetic status of the unnatural gradient restriction. To our knowledge, this is the first report of a fully unnatural gradient phonotactic restriction on segmental structure. The existence of unnatural probabilistic phonology has broad theoretical consequences because it requires probabilistic approaches to phonology to derive unnatural patterns. While weighted constraint approaches are more powerful than non-probabilistic approaches, we argue that models without unnatural constraints are not sufficient for deriving the Tarma Quechua data. We adopt a framework for evaluating phonological analyses using the goodness of fit measurement. This quantitative evaluation approach supports analyses with some,

but not all, of the relevant conceivable unnatural constraints. We argue that the proposed approach can serve to evaluate other models and approaches.

Keywords: naturalness in phonology, analysis evaluation, goodness of fit, gradient phenomena, phonotactics, maximum entropy grammar

1 Introduction

The question of how phonetic naturalness shapes language is relevant both for theories of synchronic grammar (Chomsky and Halle 1968; Stampe 1973; Hellberg 1978; Donegan and Stampe 1979; Anderson 1981; Westbury and Keating 1986; Archangeli and Pulleyblank 1994; Hayes 1999; Wilson 2006; Carpenter 2010; Becker et al. 2011; White 2013; Hayes and White 2013; Beguš 2018; Beguš and Jurgec 2026) and for a discussion on how phonetic, cognitive, and historical factors influence phonological patterns in the world’s languages (Moreton 2008; Blevins 2004; Beguš 2019, 2020, 2022). We present here a case of phonotactic restrictions on segmental features (specifically, the feature $[\pm\text{voice}]$) that are both lexically gradient (Coetzee and Pater 2008) and phonetically unnatural (Buckley 2000; Yu 2004; Coetzee and Pretorius 2010). Such cases are interesting because they suggest that these phonetically arbitrary or unnatural restrictions extend not only to categorical bans of certain sequences (Coetzee and Pater 2008), but even to manipulating the relative frequencies of sequences in the lexicon.

1.1 Background

Two questions have received increased attention in the phonological literature: (i) how to represent gradient phonotactic restrictions in the grammar (Frisch et al. 2004; Anttila 2008; Coetzee and Pater 2008; Wilson and Obdeyn 2009), and (ii) whether and how to represent unnatural processes in the grammar (Hayes 1999; Hyman 2001; Blevins 2004, 2008; Wilson 2006; Hale and Reiss 2008; Samuels 2009; Carpenter 2010; Coetzee and Pretorius 2010; Becker et al. 2011; White 2013; Hayes and White 2013; Beguš 2022). To our knowledge, however, there exists no systematic treatment of the intersection of these two topics: unnatural gradient phonotactics, i.e. phonotactic restrictions that, given a particular environment, target a single (segmental) feature and gradiently favor the value of this feature that is unnatural in that environment.

Phonetically unmotivated (see Section 1.2) phonotactics are primarily discussed in light of the “surfeit of the stimulus” problem: there exist unmotivated statistically significant trends in the lexicon, but speakers often fail to generalize these unmotivated trends to nonce words (see, for instance, Becker et al. 2011, 2012; Hayes and White 2013). Hayes and White (2013), for example, identify ten potential “unnatural” phonotactic restrictions based on systematic gaps in the English lexicon and test in a behavioral experiment whether those restrictions are internalized by native speakers. Using Hayes and Wilson (2008)’s Phonotactic Learner to identify systematic gaps, they find that, for instance, the English lexicon features a restriction against $/\mathfrak{z}/$ before a stressed vowel followed by an obstruent ($*[+\text{continuant}, +\text{voice}, -\text{anterior}]$

[+stress] [−son] in constraint formalism; Hayes and White 2013), yet learners did not generalize this to nonce words as strongly as they generalize phonetically motivated constraints. Most studies on this problem agree that speakers do not generalize unmotivated phonotactic restrictions to nonce words, with the suggestion that they might not be represented in grammar (Becker et al. 2011, 2012, 2017; Wilson and Gallagher 2018). Other studies (for instance, Hayes et al. 2009; Hayes and White 2013) suggest instead that unmotivated processes can actually be generalized to nonce words, but to a lesser extent than natural processes.

Besides its unnaturalness, the other aspect of our data that bears implications for phonological theory is the fact that restrictions are *gradient* rather than categorical. The necessity of encoding gradient phonotactics in the grammar has recently been motivated on the basis of gradient lexicalized co-occurrence restrictions (Berkley 2000; Frisch et al. 2004; Anttila 2008; Coetzee and Pater 2008; Wilson and Obdeyn 2009; Zuraw 2010) as well as other processes (Martin 2007). Coetzee and Pater (2008), for example, discuss two cases of homorganic consonant co-occurrence restrictions: Arabic and Muna. In both cases, an Obligatory Contour Principle constraint (Leben 1973; McCarthy 1986) creates a pressure that is strong, but fails to reach categorical status because of lexical variation: homorganic consonants are dispreferred within the same root, but not completely ruled out. This dispreference, however, operates in the *natural* direction: a restriction against the co-occurrence of homorganic consonants is strongly motivated by both articulatory (Garrett and Johnson 2013) and perceptual factors (Gallagher 2010).

As we will argue in Section 1.2, most cases of what the literature thus far calls “unnatural” phonotactic restrictions in the literature do not actually run counter to phonetic tendencies but rather are neutral with respect to these, including the restrictions that were tested in the *surfeit of stimulus* experiments. To distinguish our cases from these, we will adopt a new subdivision of naturalness (Beguš 2019) and focus on gradient processes that are not only phonetically unmotivated, but that operate against universal, phonetically motivated tendencies.

1.2 On naturalness

Traditionally, the naturalness of phonological processes has been conceptualized as a binary split between natural and unnatural processes (see, for instance, Kiparsky 2006, 2008). Natural processes are phonetically well-motivated and typologically frequent. Their high typological frequency is driven by the existence of phonetic precursors, which Yu (2013, 398) defines as “systematic contextually-induced variations that parallel some cross-linguistically recurring sound patterns”. Unnatural processes, on the other hand, are typologically rare or non-existent and lack phonetic precursors.

However, such a division is insufficient, as it fails to identify an important distinguishing feature within the “unnatural” group itself. All processes in the group traditionally called “unnatural” indeed lack phonetic motivation or phonetic precursors. However, a subset of these processes not only lacks phonetic motivation, but also operates against universal phonetic tendencies — that is, against phonetic precursors

responsible for a large number of (natural) phonological processes across many languages. Beguš (2019) proposes a new division of phonological processes with respect to naturalness which we follow in this paper.¹

Consistent with the literature, natural processes are defined as having clear articulatory or perceptual motivations (phonetic precursors; Yu 2013) and being typologically prevalent. Unmotivated and unnatural processes are similar in that they both lack articulatory/perceptual motivation, i.e., they have no phonetic precursors. However, in contrast to unnatural processes, unmotivated processes have no universal phonetic tendencies that militate against them — see the examples in (2), taken from Blevins (2008). On the other hand, (true) unnatural processes not only lack articulatory/perceptual motivation, but also operate against universal phonetic tendencies. In other words, if $A \rightarrow B / X$ is a universal phonetic tendency, $B \rightarrow A / X$ is an unnatural process.

To be precise, a *universal phonetic tendency* is a phonetic pressure that has a clear articulatory or perceptual motivation, and operates passively (without active articulatory control) across different languages (Beguš 2019).

- (1) *Universal Phonetic Tendency* (UPT) (Beguš 2019):
 “UPTs are phonetic pressures motivated by articulatory or perceptual mechanisms (for an overview of phonetic mechanisms, see Garrett and Johnson 2013) that passively operate in speech production cross-linguistically and result in typologically common phonological processes.”

Unlike simply “unmotivated” phonological processes, the tendencies described in this paper are “unnatural” in the sense that they not only lack phonetic motivation, but operate against universal tendencies. It is in this sense that the phenomena in this paper—such as lack of obstruent voicing post-nasally—are not only simply phonetically unmotivated but phonologically unexpected. The use of “unnatural” to describe such a process is chosen especially with reference to this latter property, following usage by Beguš (2019) and others.

For example, post-nasal voicing, intervocalic voicing, and word-final devoicing are universal phonetic tendencies: they have clear articulatory motivations, are amply attested cross-linguistically as typologically common phonological alternations, and operate passively even in languages in which such processes are not among the synchronic alternations (for discussion, see Section 2).

The literature on (un)naturalness has primarily focused on unmotivated processes. Most of the 26 listed examples in the survey of “unnatural” alternations in Blevins (2008) and the ten “unnatural” gradient phonotactic restrictions in Hayes and White (2013) are unmotivated rather than unnatural, according to the definition in Beguš (2019). Consider, for example, the following processes from Blevins (2008) and Hayes and White (2013).²

¹Similar, but slightly different distinctions have been proposed before: see Morley (2014). For Morley (2014), for example, “anti-natural” processes need to violate implicational universals and they are defined as “unattested patterns that do not conform with posited language universals”.

²Unmotivated processes are often also complex (i.e. they manipulate multiple features). Avoidance of complex alternations has long been established as complexity bias Moreton (2008). For a discussion on the complex interplay of naturalness and complexity, see Beguš (2022).

- (2) *Some processes labeled as “unnatural” in Blevins (2008) and Hayes and White (2013)*
- a. /p/ → [s] / __i
 - b. /i/ → [u] / d__
 - c. * $\begin{bmatrix} +COR \\ +cont \\ -strid \end{bmatrix} \begin{bmatrix} -stress \\ +round \end{bmatrix}$ (“No [θ, ð] before stressless rounded vowels”)
 - d. * $\begin{bmatrix} +cont \\ +voice \\ -ant \end{bmatrix} \begin{bmatrix} +stress \\ -son \end{bmatrix}$ (“No [ʒ] before stressed vowel + obstruent”)

All these processes lack phonetic motivation, but they do not operate against universal phonetic tendencies. In other words, it is not the case that [θ, ð] in the context before unstressed rounded vowels are universally preferred: there exists no cross-linguistic passive phonetic tendency that would prefer fricatives before unstressed rounded vowels.

Some gradient restrictions that operate in the unnatural direction have recently been reported in light of the surfeit of the stimulus experiments mentioned in Section 1.2, including (i) a higher rate of alternation in monosyllabic compared to polysyllabic words (Becker et al. 2012), and (ii) a stronger preference for antepenultimate stress in LLL words compared to HLL words (Garcia 2017). Strictly speaking, the first restriction does not contradict any universal phonetic tendency (perhaps just a universal phonological tendency). The second restriction targets non-segmental features and while the restriction is significant, it is also very subtle in magnitude (see Garcia 2017).

Few truly unnatural active alternations are reported (for an overview, see Beguš 2018; Dabkowski and Beguš 2024; Beguš and Dabkowski 2025); there is additional debate as to whether unnatural processes are, in fact, possible as productive synchronic alternations. The most compelling case of an unnatural synchronic alternation is post-nasal devoicing (PND) in Tswana and Shekgalagari (Hyman 2001). PND qualifies as unnatural according to all criteria laid out in this section: it operates against the well-motivated, typologically common, and passive phonetic tendency of post-nasal voicing. Coetzee and Pretorius (2010) have shown that PND is phonetically real and fully productive: speakers generalize it to nonce words (although it does not appear in all varieties and it might have alternative analyses; Gouskova et al. 2011; Downing and Hamann 2021). On the other hand, one of the most convincing cases against unnatural alternations in synchronic grammars is that of final (de)voicing. While final devoicing is a highly common and phonetically motivated process (Steriade 1997; Iverson and Salmons 2011) its unnatural counterpart — final voicing — is not attested in any language as a synchronic process (Kiparsky 2006, 2008; for a possible exception, see Yu 2004; de Lacy 2002; Blevins et al. 2020). This has led some scholars to conclude that final voicing is an impossible synchronic alternation (Kiparsky 2006, 2008). The question of whether fully productive synchronic alternations can be unnatural remains open.

1.3 Aims

To our knowledge, no systematic treatment of unnatural gradient phonotactic restrictions exists in the literature. This paper presents a novel analysis of the lexicon of the Tarma dialect of Quechua (data from Adelaar 1977; Puente Baldoceda 1977 and novel unpublished data) which has a highly unnatural trend in the lexicon that targets a single segmental laryngeal feature $[\pm\text{voice}]$ at a considerably greater magnitude than any cases of unnatural gradient restriction reported thus far. In our reported case, the distributions are approaching categorical levels (for example in the post-nasal and post-obstruent positions). We show that these trends run counter to specific universal phonetic tendencies, and argue that the trends are statistically significant, phonetically real, and morphophonologically productive. Based on these findings, we interpret unnatural trends as a gradient phonotactic restriction (in line with Coetzee and Pater 2008 and others).

The existence of unnatural gradient phonotactic restrictions has several implications. One of the most consequential debates in phonology is to what degree it is influenced by phonetics and whether it can accommodate unnatural processes (Hyman 2001; Hayes and Steriade 2004; Blevins 2004; Kiparsky 2006, 2008; Reiss 2018; Beguš 2022). Our data speak to this debate on two fronts: first, we argue that phonotactic restrictions can extend in phonetically unnatural directions. Second, phonological processes are not only categorical, but often operate probabilistically. Our presented case suggests that not only categorical, but also gradient phonological processes can operate in the unnatural direction.

The analysis in this paper raises a theoretical question for the theory of Markedness (see, e.g., de Lacy 2002, 2004, 2006; de Lacy and Kingston 2013) in weighted-constraint theories like Harmonic Grammar or MaxEnt (Goldwater and Johnson 2003; Coetzee and Pater 2008, 2011; Pater 2008, 2009; Albright 2009; Potts et al. 2010). While weighted constraint models are able to derive gradient phonotactics (see, e.g., Coetzee and Pater 2008), we show that, without unnatural Markedness constraints, they are not always able to derive phonotactic systems in which the unnatural element in that context is more frequent than the natural element in a given context. We call this effect the “Natural Gradience Bias”.³ Since the cases presented here are of this exact nature, this opens up the possibility that Markedness constraints might, after all, be able to counter universal phonetic tendencies (cf. Zhang 2010 and other work in that direction).

We also apply a model comparison paradigm to investigate the necessity of unnatural constraints to model our data in a grammar framework. To our knowledge, explicit formal model comparison has not been applied before to the natural/unnatural debate. We model Tarma Quechua data with natural and selected unnatural constraints, then compare the goodness of fit of various models (e.g., Hayes et al. 2012). Such simulations, as well as the proposed evaluation technique, bring several advantages. Firstly, we can test whether phonetically unnatural gradient restrictions can be derived with natural constraints only. Secondly, we can quantitatively evaluate different analyses

³In classic OT, the connection between natural constraints and corresponding patterns is straightforward, but MaxEnt does not feature harmonic bounding (e.g. Hayes 2016) and weighted constraint theories in general allow cumulative constraint interaction (Potts et al. 2010), so that candidates not preferred by any single constraint may still surface.

	[+voice] preferred	[+voice] dispreferred
a. Initial devoicing		word-initially
b. Intervocalic voicing	between two vowels	
c. Post-nasal voicing	after [+nasal] consonant	
d. Voicing agreement	adjacent to [+voice] consonant	adjacent to [-voice] consonant
e. Coda/final devoicing		word-finally and in codas

Table 1 Some universal phonetic tendencies for [+voice].

and argue in favor of unnatural constraints based on a quantitative metric. Finally, the technique also facilitates interpretability of MaxEnt models: we provide explanations for various predictions made by the MaxEnt simulations.

2 Universal tendencies for voicing

The case of unnatural phonotactics presented in this paper targets a single phonological feature: $[\pm\text{voice}]$. Alternations and phonotactic restrictions targeting this feature are among the most well-studied phenomena in phonology and we have a substantial body of research on the typology as well as the phonetics of voicing cross-linguistically. The attention that voicing has received in the literature helps us determine in which environments the voice feature is universally (dis)preferred, i.e. which processes targeting $[\pm\text{voice}]$ are “unnatural” according to the definition above (see (1)). Table 1 summarizes these tendencies.

A body of research has established that voicing is universally aerodynamically dispreferred in word-initial and word-final or coda positions (Westbury and Keating 1986; Iverson and Salmons 2011); it is universally preferred intervocalically (Westbury and Keating 1986; Davidson 2016) and after a nasal (Rothenberg 1968; Kent and Moll 1969; Ohala and Ohala 1993; Ohala 1983; Hayes and Stivers 2000). Voicing is also dispreferred in clusters and geminates: the longer the closure, the more antagonistic aerodynamic forces are to voicing (Westbury and Keating 1986). Finally, the voicing feature is also universally dispreferred before another obstruent with a different value in voicing: clusters that disagree in $[\pm\text{voice}]$ are universally dispreferred (Myers 2010).

Tarma Quechua phonotactic restrictions violate naturalness in two main contexts: post-nasally and in consonant clusters. post-nasal voicing fulfills all three requirements for a universal phonetic tendency from section 1.2. To begin, it is a common productive phonological alternation and sound change (Kümmel 2007): as reported in Hayes and Stivers (2000), Locke (1983)’s study identifies 15 out of 197 (or 8%) languages surveyed with post-nasal voicing as a synchronic alternation. The articulatory explanation for post-nasal voicing is described in detail in Hayes and Stivers (2000) and Coetzee and Pretorius (2010), building on previous work by Rothenberg (1968), Kent and Moll (1969), Ohala (1983), Ohala (1993), and others. Two basic factors are identified: in the transition from the nasal stop into the oral stop of nasal-plosive clusters, the velum has to rise from its lowest position to a complete closure. During this transition, air can still escape through the nasal cavity (“nasal leakage”), which makes it more difficult to achieve the increase in supraglottal pressure necessary to cease

voicing (cf. Coetzee and Pretorius 2010). Secondly, as the velum rises, the volume of the oral cavity increases. This also promotes voicing, as the increased oral cavity volume translates to more time where voicing is possible (Hayes and Stivers 2000; Coetzee and Pretorius 2010). post-nasal voicing is also well-motivated from a perceptual perspective: cues for voicelessness, such as the release burst of a stop, or cues for voicing, such as low frequency energy, are reduced in post-nasal position (Coetzee and Pretorius 2010, p. 405). Finally, voicing is a passive phonetic tendency in post-nasal position, i.e. stops universally feature more voicing into closure post-nasally than in other positions (Hayes and Stivers 2000; Davidson 2016).

The second universal phonetic tendency that Tarma Quechua’s phonotactics operate against is voicing agreement in obstruent clusters, which, again, fulfills all three conditions mentioned above. Clusters that agree in voicing are typologically very common and voicing assimilation is one of the most common processes cross-linguistically (Myers 2010). Myers (2010) lists at least 28 such languages; this list is neither exhaustive, nor does it result from a survey. The phonetic motivation is straightforward: laryngeal features “overlap and blend” in obstruent clusters, which results in the passive voicing of preceding voiceless stops due to laryngeal coarticulation (Myers 2010, 164ff). Perceptual factors have also been proposed to promote the agreement of laryngeal features (Myers 2010). Finally, agreement in laryngeal features fulfills the third condition of a universal phonetic tendency: voicing before another voiced stop is a passive phonetic tendency (Barry and Teifour 1999) and is attested in languages in which voicing assimilation is not a complete synchronic process. As (Myers 2010, p. 164) claims, “[a]coustic studies have shown that there is a longer voiced interval in an obstruent before a voiced consonant than before a voiceless consonant in English (Haggard 1978; Docherty 1992; Smith 1997; Jansen 2004), French (Snoeren et al. 2006), and Syrian Arabic (Barry and Teifour 1999)”.

3 Unnatural trends in the lexicon

This section shows that the identified trends in the lexicons that operate against the two universal phonetic tendencies are statistically significant, phonetically real, and point to evidence in favor of the limited morphological productivity of these processes.

3.1 Tarma Quechua

Tarma Quechua is a dialect of Quechua (Quechua I) spoken in the Tarma district of the Junín province of Peru. The number of speakers is difficult to establish, as the dialect is rapidly being replaced by Spanish (Adelaar 1977). Adelaar (1977) and Puente Baldoceda (1977) report approximately 30,000–40,000 inhabitants of the Tarma district and an additional 3,500 inhabitants of La Unión Leticia. However, the number of speakers of the dialect with the particular unnatural phonotactics that are of interest here is much smaller and difficult to estimate. The unnatural phonotactic restriction presented here is also reported in the dialect of Paccho (Adelaar and Muysken 2004). Because there are no descriptions or recordings of Paccho Quechua available, we leave this dialect out of our discussion.

The segmental inventory of Tarma Quechua includes 23 consonant and 10 vowel segments. This includes four voiceless stops (/p/, /t/, /k/, /q/), three voiced stops (/b/, /d/, /g/), four voiceless fricatives (/f/, /s/, /ʃ/, /x/), two voiceless affricates (/tʃ/, /tʂ/), four nasals (/m/, /n/, /ɲ/, /ŋ/), four liquids (/l/, /ʎ/, /r/, /ɾ/), and two semivowels/glides (/w/, /j/).⁴ The vowel inventory consists of long/short pairs of /a/, /i/, /e/, /o/, and /u/ (Adelaar 1977).

3.1.1 Stop voicing

The Quechua dialect continuum almost uniformly has only voiceless stops in native vocabulary (Adelaar and Muysken 2004). Adelaar (1977) and Puente Baldoceda (1977) report that some of these voiceless stops have become voiced in Tarma Quechua (henceforth: TQ), or more precisely, the Quechua dialects spoken in Tarma, Huaricolca, Palcamayo, and La Unión Leticia. Voiceless velar and labial stops (to the exclusion of alveolars) are reported to undergo voicing in intervocalic and post-consonantal positions, but not after a nasal consonant. Adelaar (1977) and Puente Baldoceda (1977) note that voicing does not apply categorically (and that it does not apply post-nasally), but no further analyses on the lexicon are performed. Below we present results of a statistical analysis of the TQ lexicon that reveals a highly unnatural lexical trend. We show that, in addition to limitations on voicing after nasals, the relative rates of voicing in different environments, including intervocalic and post-consonantal positions, contradict the universal phonetic tendencies from Section 2. This paper, to our knowledge, is the first report of this unnatural distribution in TQ.

The obstruent voicing observed in this paper is contrastive in the described environments, but has a limited functional load. It is indeed true that some voiced/voiceless obstruent minimal pairs exist in Tarma Quechua (e.g. *tʃimba-* ‘to put a bag on someone’s back [colocar un bulto en la espalda de una persona]’ and *tʃimpa* ‘opposite side (of a watercourse) [orilla contraria]’), despite a historical lack of voiced obstruents. However, the phenomenon discussed here is that of gradient lexical phonotactics: the preference within the lexicon for phonetically dispreferred outcomes. Thus, the question is not whether voiced and voiceless obstruents form distinct phonemes in Tarma Quechua, but rather why the phonemes within Tarma Quechua as a whole are distributed in a phonetically unnatural way.

3.2 Data

For the purpose of the analysis, we collected all tokens of prevocalic labial and velar stops from the vocabulary list in Adelaar (1977) and an unpublished lexicon by Puente Baldoceda. Altogether 2,097 instances of target consonants (labial/velar stops, see section 3.3.1) across 1,679 words were thus collected. As a comparison, Ernestus and Baayen (2003) analyze 1,697 entries from the CELEX database; Hayes and Londe (2006) use 10,974 unique stems for their analyses. The entries were annotated for place of articulation (labial or velar), voicing, part of speech, loanword status, derived status, left and right context, source dictionary, and their position in the word (in terms

⁴The consonants shown here as /ʃ, tʃ/ are transcribed as <š, č> but characterized as palatal in Adelaar (1977). The sound shown as /ɾ/ in practice alternates between [z, r].

	#	N	V	R	S	T
voiced	14	11	158	82	71	29
voiceless	417	83	202	28	20	3
% voiced	3.2	11.7	43.9	74.5	78.0	90.6

Table 2 Voiced vs. voiceless labial and velar stops in Tarma Quechua native vocabulary across contexts.

of syllable count). The root structure of Quechua is predominantly disyllabic. Some entries in our data include words analyzed as non-derived that involve more than two syllables (Muysken 2011; Gouskova 2023). The question of whether a lexical item is analyzed as derived or non-derived by the speakers is non-trivial. In order to avoid subjective analyses of the derived status of lexical entries, we follow the analysis of roots in the two dictionaries. While it is possible that some entries marked as non-derived might be analyzed as derived by the speakers, we do not expect these mismatches to alter the results substantially. For details on data collection and annotation, see Section A.1 in the Supplementary Materials.

3.3 Statistical analysis

3.3.1 Individual segments

The raw data analysis of non-loan non-derived vocabulary in Table 2 reveals a surprising trend: voicing surfaces very rarely post-nasally (11.7%) (e.g. *ayku* vs. *saygu*), in almost half of the lexicon intervocalically (43.9%) (e.g. *a:pa*, *aba*), and almost always post-consonantly, including in positions after a voiceless fricative (78.0%) (e.g. *alisga* vs. *alajmoska*) or voiceless stop (90.6%) (e.g. *pakba* vs. *pikpif*).⁵ Raw counts suggest that voiced stops are least frequent word-initially and post-nasally, and most frequent after voiceless obstruents. To test the statistical significance of this trend, we fit a logistic regression model to the data with the R statistical software (R Core Team 2018) using the *glm()* function. The dependent variable was binary: presence or absence of voicing (voicing coded as success); the independent variables were PLACE of articulation and LEFT CONDITION (i.e., the phonological class of the immediately preceding segment; additionally, when Left Condition = V, only intervocalic segments are selected). The model also includes the PART OF SPEECH predictor (with four levels; nouns, verbs, dual noun/verbs, and other) to control for potential effects of parts of speech on the voicing pattern and the DICTIONARY source predictor (to test whether the described voicing patterns differ across the two dictionaries).

The final model (AIC = 906.07) includes PLACE (sum-coded with two levels: labial and velar)⁶ and LEFT CONDITION (treatment-coded with six levels: initial, post-nasal, intervocalic, post-sonorant, and post-fricative and post-stop, with intervocalic as the reference level). While their interaction is not warranted by AIC, we leave it in the model in order to illustrate the same trends in voicing across the two places of

⁵Unless noted otherwise, we will henceforth abbreviate phonological classes as follows: T – voiceless stop, D – voiced stop, S – voiceless fricative, N – nasal, R – non-nasal sonorant, including glides, V – vowel.

⁶For a discussion on contrast coding, see Brehm and Alday (2022). For a discussion on how the final model was chosen, see Section A.2 in the Appendix.

articulation. The significance of the estimates of interest does not differ between the best-fitting model (which includes the `DICTIONARY` predictor) and the final model (with only Place, Left Condition, and their interactions) or between models with or without the Place:Left Condition interaction.

The estimates of the final model in Figure 1 and Table 3 illustrate the unnatural trends in the Tarma Quechua lexicon. In the intervocalic position (V_V), voiced and voiceless stops are nearly equally frequent ($\beta = -0.18, z = -1.62, p = 0.11$). Voiced stops are significantly less frequent word-initially ($\beta = -3.40, z = -9.79, p < 0.0001$) and post-nasally than intervocalically ($\beta = -1.96, z = -4.61, p < 0.0001$). On the other hand, voiced stops are significantly more frequent after non-nasal sonorants and glides ($\beta = 1.52, z = 5.20, p < 0.0001$), after voiceless fricatives ($\beta = 1.67, z = 5.16, p < 0.0001$), and after voiceless stops ($\beta = 2.49, z = 3.83, p = 0.0001$) than they are intervocalically. No overdispersion was detected in the models (ratio = 1.01; tested with *overdisp_fun()* in Bolker 2019).

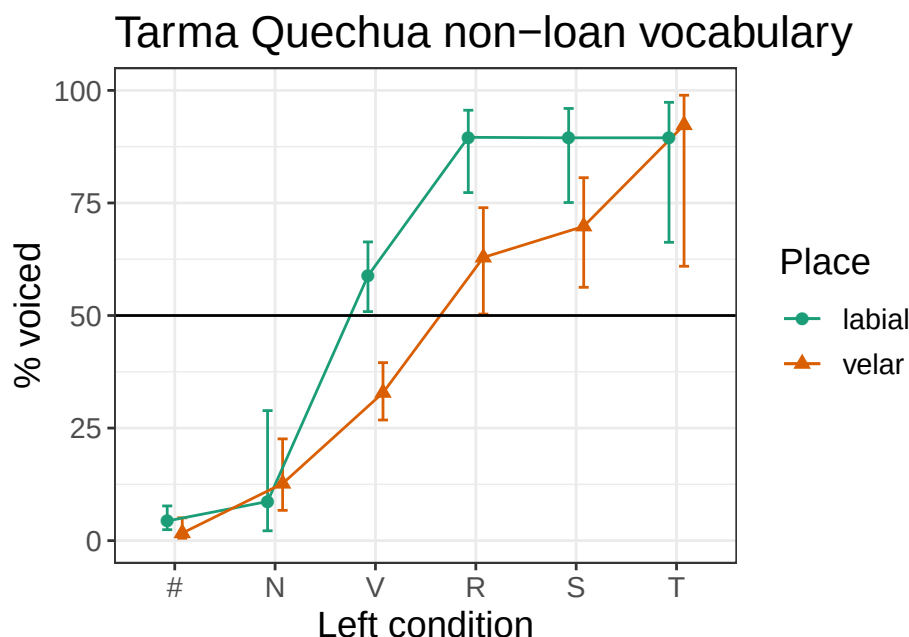


Fig. 1 Percentage of voiced stops according to position based on a logistic regression model in Table 3. All bars in figures represent 95% confidence intervals based on model estimates.

These trends are significant even if we include loanwords and derived words in the analysis. Data with the entire collected vocabulary including loanwords and derived words (total $N=1,793$) were fit to the same model with `LEFT CONDITION`, `PLACE` and their interaction as predictors. Estimates in Table A1 in the Supplementary Materials (A.2) suggest all differences remain significant in this model.⁷

⁷The interaction is significant in this model.

	Estimate	Std. Error	z value	Pr(> z)
(Intcpt)=V	-0.1791	0.1105	-1.62	0.1051
#	-3.4002	0.3474	-9.79	0.0001
N	-1.9615	0.4254	-4.61	0.0001
R	1.5191	0.2921	5.20	0.0001
S	1.6683	0.3232	5.16	0.0001
T	2.4916	0.6502	3.83	0.0001
Place1	0.5358	0.1105	4.85	0.0001
#:Place1	-0.0433	0.3474	-0.12	0.9007
N:Place1	-0.7466	0.4254	-1.76	0.0792
R:Place1	0.2760	0.2921	0.95	0.3447
S:Place1	0.1150	0.3232	0.36	0.7219
T:Place1	-0.7082	0.6502	-1.09	0.2760

Table 3 Logistic regression model with PLACE (sum-coded) and LEFT CONDITION and their interaction as predictors (non-loan non-derived vocabulary)

Furthermore, if we add LOANWORD status as a predictor to our model (including loan and derived vocabulary), the fit improves significantly, but it is not clear whether loanword status as a predictor is justifiable in a cognitive model. If speakers are aware of the loanword status of the vocabulary, the distribution of $[\pm\text{voice}]$ for loanwords ceases to be of interest (in this case speakers may use two different grammars that govern native and loanword phonologies; see, e.g., Itô and Mester (2002); Itô and Mester (2003)). The model with three independent variables was fit to the data: LEFT CONDITION, PLACE of articulation, and LOANWORD status (sum-coded). Two two-way interactions are significant (LEFT CONDITION:PLACE and LEFT CONDITION:LOAN) and are included in the model. The significance of all main effects remains the same as before (at means of other predictors), except that the V__V vs. S__ and V__V vs. T__ cease to be significant as a main effect, likely due to data scarcity. There is only one loanword with labials or velars in post-stop position, and only twelve in post-fricative position. Other frequency differences remain significant: there is still more voicing intervocally than post-nasally.

If we isolate loanwords from the native vocabulary, we do not observe any unnatural patterns, which is not surprising as the donor language, Spanish, does not feature any of the unnatural patterns in TQ. In a model that excludes derived words, but includes loanwords with three predictors (LEFT CONDITION, PLACE, and LOAN with all interactions), the loanword vocabulary does not reflect the unnatural pattern from the non-loan vocabulary (see Figure A2). Adding loanword status as a predictor also introduces a problem of data scarcity, since there are only a few loanwords with labials or velars in post-obstruent position. As a whole, loanwords prove inconclusive for our analysis.

To test the effect of syllable structure, we fit a model with LEFT CONDITION and SYLLABLE STRUCTURE as predictors with their interaction (on non-loan non-derived vocabulary). Controlling for the syllable structure is important because stressed, secondary stressed, and unstressed syllables could affect voicing. Given the static nature

Universal tendencies for [+voice]	Observed significant trends in TQ
T/S__ < V__V	V__V < T/S__
T/S__ < N__	N__ < V__V < T/S__

Table 4 Unnatural distribution of [+voice]

of TQ’s stress system, syllable structure is a proxy for controlling for the effects of various stress conditions. Adding SYLLABLE STRUCTURE introduces highly sparse data in some conditions, which is why it is not added to the main model and which is why we do not include PLACE as a predictor here. We divide the data into four groups according to the syllable structure: onset of the ultimate syllable in disyllabic words (the most common position given TQ root structure), onset of the ultimate syllable in trisyllabic words, onset of the penultimate syllable in trisyllabic words, and all other syllabic positions (elsewhere). The estimates of the model in Figure A3 suggest that in the ultimate syllable of disyllabic words, all trends are as observed in the full model. Occurrences of labial and velar stops in trisyllabic words are rather rare, and the effects are not significant. Raw data, however, suggest that the same tendencies do also operate in trisyllabic words. For example, in the post-nasal position in trisyllabic words, 13 (76.5%) stops are voiceless and 4 are voiced (23.5%). In the position after a voiceless stop (T), 2 stops are voiced (66.7%) and 1 is voiceless (33.3%). A notable deviation from the general trend is that in trisyllabic words, stops in a position after a voiceless fricative are voiceless in 5 cases and voiced in 1 case. In sum, the observed unnatural differences in voicing are confirmed when they all appear in the most common position given TQ’s syllable structure: the ultimate syllable of disyllabic words. In trisyllabic words, the data are sparse, but at least the distinction in voicing between post-nasal and post-stop positions operates in the expected unnatural direction.

As shown in this analysis, [+voice] in labial and velar stops is significantly less frequent word-initially and post-nasally than in intervocalic position. [+voice] is significantly more frequent in post-sonorant and post-obstruent position compared to intervocalic position in the TQ native vocabulary. These trends hold across parts of speech, syllable structures, and source dictionaries.

As argued in Section 2, post-nasal and intervocalic positions universally prefer voicing, while voiced stops after voiceless obstruents are universally dispreferred. The fact that TQ exhibits less voicing post-nasally and intervocalically than after voiceless obstruents⁸ is thus highly unnatural. TQ voicing thus operates in a direction opposite to two universal phonetic tendencies: it operates more frequently where it is universally dispreferred (post-consonantly) and less frequently where it is universally preferred (post-nasally and intervocalically). These findings are summarized in Table 4.

3.3.2 Disagreement of voicing in clusters

Another aspect of gradient unnaturalness emerges in TQ if we look into the within-context distribution of voicing in consonant clusters involving obstruents: obstruent

⁸The analysis shows post-nasal < intervocalic and intervocalic < post-obstruent, from which post-nasal < post-obstruent can be derived by transitivity.

1 st member	2 nd member	
	Labial	Velar
t	lutbi	mutgi
tʃ	/	atʃga
tʂ	atʂba	matʂga
k	takba	/
s	tʃasbu	tʂasgi
ʃ	kafbi	ifgi
x	saxbi	manexax-gunax
l	tʃilbi	tʃilgi
r	karba	argu
j	ajba	ajga
w	kawbu	awgis

Table 5 Consonant clusters by the first member with an example for each combination in TQ (from Adelaar 1977).

clusters that agree in voicing are gradiently dispreferred in TQ — clusters that disagree in voicing are significantly more frequent.

In TQ, labial and velar stops surface as voiced after a non-nasal consonant (Table 2). The following consonants are attested immediately before [b, g]: [t, tʃ, tʂ, k, s, ʃ, x, l, r, j, w]. The list includes voiceless fricatives, affricates, and even voiceless stops. The following clusters of two stops are attested: [kb, tb, tg]. Table 5 presents examples of clusters that disagree in voicing after each consonant (data from Adelaar 1977).

A statistical analysis of this trend reveals that obstruent clusters that disagree in voicing are much more frequent than clusters that agree in voicing if the second consonant is either a labial or a velar stop. Table 6 shows the number of occurrences of obstruent clusters (both stops and fricatives; for the purposes of this particular analysis, we mark all voiceless obstruents with T, and all voiced obstruents with D) in which the first element is an obstruent and the second element is a labial or a velar stop in the non-derived non-loan TQ vocabulary. To test the statistical significance of this distribution, a logistic regression model was fit to data with voicing as the dependent variable, PLACE of articulation and CLUSTER POSITION (first vs. second) without the interaction (which is not significant). In the second position in obstruent clusters (after the first obstruent which in all but two cases is voiceless), voicing is significantly more frequent than non-voicing ($\beta = 2.25, z = 5.21, p < 0.00001$). Voicing is significantly less frequent when the obstruent is in the first position compared to the second position ($\beta = -5.48, z = -6.91, p < 0.00001$). Tested differently, the proportion of clusters that disagree in voicing is significantly higher than 50% (according to the Exact binomial test; $p < 0.00001$).

TQ thus features a statistically significant trend that restricts clusters agreeing in voicing in favor of disagreeing clusters. This trend is both gradient and unnatural.

The trend against agreeing obstruent clusters in TQ is unnatural in two additional respects. Table 6 shows a preference for TD clusters, compared to DT clusters — which goes against yet another phonetic tendency. Voicing is articulatorily easier to maintain in the initial parts of closure than it is to initiate voicing after a period

	TT	TD	DT	DD
Count	23	100	2	0
Percent	18.4%	80.0%	1.6%	0%

Table 6 Voice feature in obstruent clusters in which the first element is an obstruent and the second element is a labial or a velar stop in non-derived non-loan TQ vocabulary.

of voiceless closure (Ohala and Riordan 1979; Ohala 1997). The reason for this articulatory dispreference is straightforward and has been identified as the Aerodynamic Voicing Constraint: airflow and a subglottal-supraglottal pressure difference, necessary for voicing, are sufficient during vowel articulation, but decrease into closure. The reason why voicing is articulatorily difficult to initiate after a period of voiceless closure is that it is difficult to reinstantiate the necessary airflow and pressure difference — once the closure has caused them to decrease — without releasing the stop closure completely. In addition, there is a typological tendency towards respecting the Syllable Contact Law (Vennemann 1988), which states that the coda of the preceding syllable should be more sonorous than the onset of the following syllable and which also prefers DT over TD clusters. Decreasing phonation into closure is observed as a passive tendency in several languages (see, for instance, Möbius 2004; Davidson 2016). In other words, voicing has a universal tendency to decrease rather than increase during the closure. The restriction in TQ against DT (decreasing in voicing) clusters in favor of TD clusters (increasing in voicing) is thus unnatural: it operates against the universal phonetic tendency that decreases voicing into closure. Finally, the fact that labials and velars have voiced counterparts in TQ, but alveolars do not, also goes against the phonetic tendency whereby labials are most likely to be voiced and velars least likely (Everett 2018).

3.4 Phonetics

No previous detailed phonetic analyses of the system of voicing in TQ exist: Adelaar (1977) and Puente Baldoceda (1977) are based on qualitative descriptions of recordings and are not supported by acoustic analyses. Our analysis supports the phonetic reality of the TQ voice system as described above, making the case for true unnaturalness in the TQ data. In this analysis, a segment is considered voiced if it has a clear voicing bar in the spectrogram.

The analyzed recordings were obtained online⁹ in .wav format, sampled at 90 kHz¹⁰ with 16-bit quantization and analyzed with Willem Adelaar’s permission in the Praat software (Boersma and Weenink 2015). The recordings were made by Willem Adelaar in 1970 in Tarma, in the Junín province of Peru. The informant was a 35-year old male speaker of TQ. The recordings are noisy with considerable reverberation (for a detailed acoustic analysis of reverberation, see Section A.3 in the Supplementary

⁹ Accessible online at: <https://hdl.handle.net/1839/00-0000-0000-0016-A448-A>

¹⁰ Although not specified by the archive from which the audio was retrieved, it can be assumed from their age that the recordings were originally analogue, and therefore have no original sampling frequency.

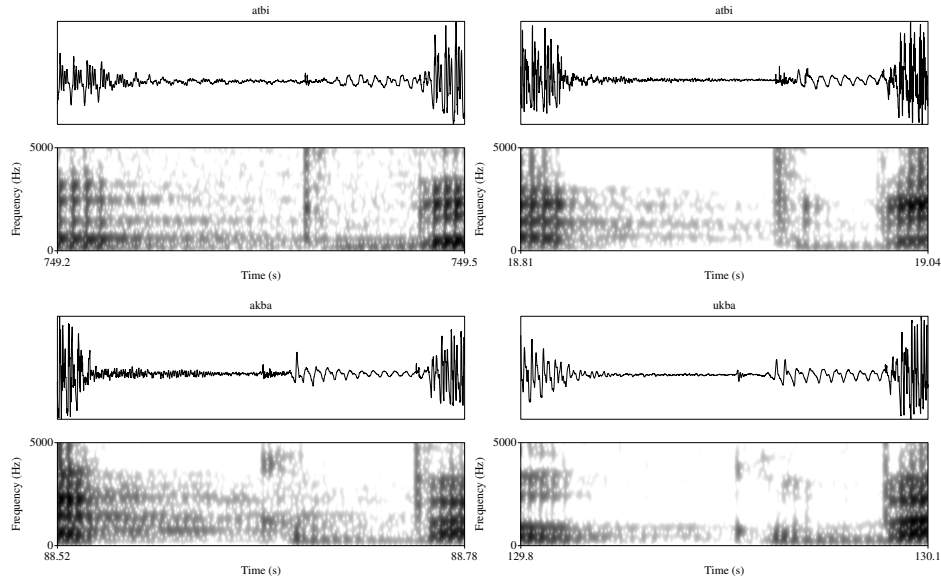


Fig. 2 Waveforms and spectrograms of four TD clusters: [atbi], [atbi], [akba], and [ukba].

Materials), but the analysis nevertheless reveals important aspects of the unnatural gradient phonotactics and of the phonetic system of TQ in general. The following discussion includes waveforms and spectra that were chosen as best representing the described voicing pattern. Given the quality of the recordings and a substantial presence of echo noise, a more comprehensive quantitative phonetic analysis is difficult to undertake.

Figure 2 shows four waveforms and spectrograms of two TD clusters: [tb] and [kb]. Around the middle of each of the clusters, we see the release burst of the first consonant ([t] or [k]), signified by an abrupt and very short burst of energy in the spectrogram. In the portion before this burst, all four spectrograms show a voiceless plosive with almost no phonation into closure, while the portion after the burst shows clearly expressed phonation during closure (i.e., a truly pre-voiced second consonant).

The same situation holds when the first element is an obstruent other than a stop, such as in the [sb] or [sg] sequences in Figure 3. The lack of any low frequency energy in the fricative portion of the cluster confirms that the first element, [s], is voiceless. Phonation starts at the onset of the stop.

The exact realization of voiced stops in clusters may vary. A precise distribution is difficult to establish with limited data, but a short transitional vocalic element is occasionally found between the voiceless and voiced obstruent, indicating a smaller degree of gestural overlap (Hall 2011; Figure 5).¹¹ Occasionally, the voiced element surfaces as a continuant (though whether it is an approximant or fricative is harder to distinguish). These occur in apparent free variation. Figure 4 presents waveforms and spectrograms of phonemic voiced stops that surface as voiced continuants, as indicated

¹¹Occasionally, the second element is found to surface as voiceless or deleted.

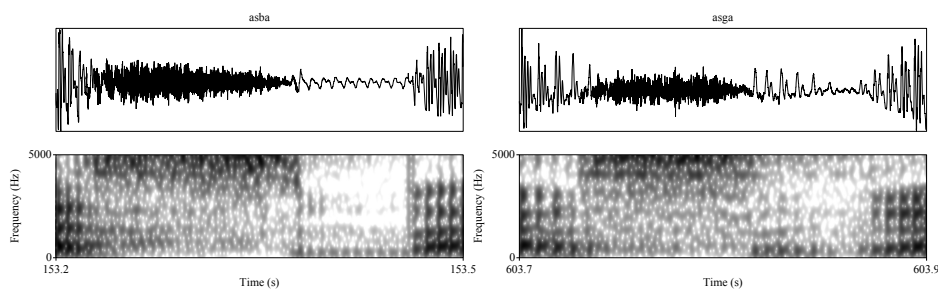


Fig. 3 Waveforms and spectrograms of two TD clusters: [sb] and [sg].

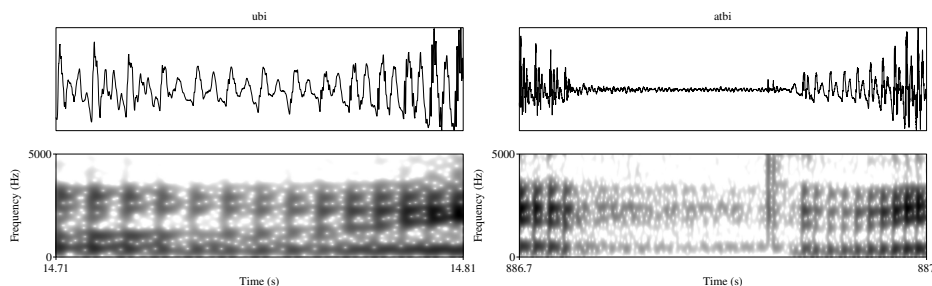


Fig. 4 Waveforms and spectrograms of voiced stops that surface as voiced continuants: intervocally (left) and after a voiceless stop (right).

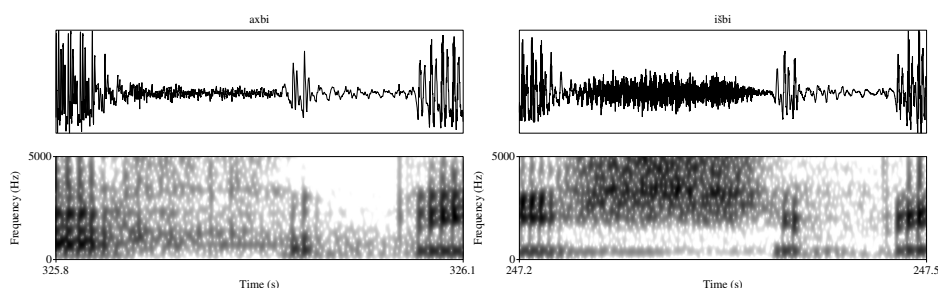


Fig. 5 Waveforms and spectrograms of voiced labial stops in post-consonantal position with a short vocalic element between the voiceless and voiced elements: [xb] (left) and [fb] (right).

by the presence of formant structure during the closure. These formants are present throughout the consonantal part, even in cases where formants do not result from reverberation (see Section A.3 in the Supplementary Materials) — after a voiceless stop.

After nasals, on the other hand, voiceless stops are the preferred variant, as detailed in Table 3. Figure 6 shows spectrograms with voiceless [p] and [k] after nasals. Also note that voiceless stops in TQ are unaspirated, which means that the phonotactic restriction in fact targets the feature $[\pm\text{voice}]$ rather than the feature $[\pm\text{spread glottis}]$.

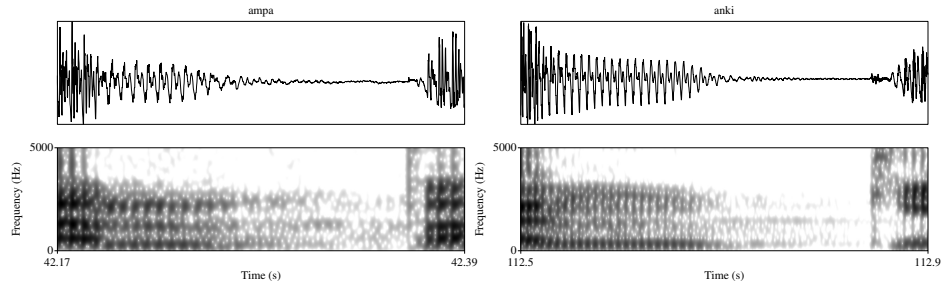


Fig. 6 Waveforms and spectrograms of voiceless stops in post-nasal position: [mp] (left) and [ŋk] (right).

3.5 Morphological alternations

Corroborating its status as a phonotactic restriction, there exists evidence that the gradient phonotactic restriction is synchronically active in some morphophonological alternations. Creider (1968) and Adelaar (1977) identify four suffixes with an initial voiced labial stop that feature morphophonemic alternation:

- (3) *Alternating suffixes*
- a. -ba/-pa ‘genitive’
 - b. -bax/-pax ‘purposive’
 - c. -bita/-pita ‘procedentive’
 - d. -bis/-pis ‘even, too’

The allomorph with voiced initial stops is selected after vowels and non-nasal consonants, including voiceless obstruents; the allomorph with voiceless initial stop is selected after nasals (Creider 1968). The distribution is illustrated in (4).

- (4) a. *Intervocalic*
wawxi-gi-**ba** waji-n
‘the house of your brother’
- b. *post-nasal*
waji-n-**pa** pasa-fun
‘we’re going to walk by way of his house’
- c. *Post-obstruent*
tamja-ja-n nuqa-ntʃik-**baq**
‘it is raining now for us’ (Creider 1968, pp. 12-13)

This process is productive for a subset of suffixes. Other suffixes do not enter the alternation. For example, the highly frequent plural suffix /-guna/ and other suffixes /-bura/, /-gama/, and /-gasga/ have no voiceless allomorphs in post-nasal position (Adelaar 1977, p. 59). The productivity of this morphophonemic alternation also differs across the dialects. (Adelaar 1977, p. 59) reports that voiceless allomorphs are required in Vicora Congas, whereas in Huanuquillo the rate of application varies, i.e. is gradient.

There exists a possibility that morphophonological alternations in (4) are lexicalized or governed by non-phonological parts of grammar and that speakers would not

productively extend the distributions to wug-tests. The possibility that a phonological process is not productive is always present when studies do not perform wug-tests (which represents the majority of phonological literature). Because further fieldwork was not possible, we leave experimental tests of productivity for future research.

4 Grammar learning simulations

The results from the statistical and acoustic analyses in Section 3 show both phonetically motivated and phonetically unnatural gradient patterns in TQ. In this section, we investigate how these map onto constraints within a Maximum Entropy (MaxEnt) framework (Goldwater and Johnson 2003; Hayes and Wilson 2008).

We assume here that the grammar reflects the patterns described in Section 3, as often happens with quantitative patterns in phonology (e.g., Zuraw 2010). There is the possibility that the patterns are completely non-productive and we are dealing with “surfeit of the stimulus” (Becker et al. 2011, 2012), in which case we must account for their non-productivity separately. In both cases, it is important to know which constraint set could account for the observed patterns.

Whether phonetically unnatural constraints are needed in the grammar has been a long-standing discussion in phonology (Hyman 2001). In this section, we approach this question by simulating grammars with natural and unnatural weighted constraints. We use model selection techniques to investigate which natural/unnatural constraints are necessary for deriving gradient phonotactic restrictions. In doing this, we demonstrate the existence of a “natural gradient bias” in MaxEnt grammars: the presence of only natural constraints cannot motivate all unnatural gradient processes, *pace* Magri and Anttila (2018).¹² Such a bias is to be expected: natural constraints prefer natural configurations, so that zero weight on any natural constraint gives unnatural and natural candidates equal probabilities, and non-zero weight would give natural candidates a greater probability than unnatural ones, so that unnatural candidates should never prevail over natural ones. Our approach also facilitates interpretation of what various MaxEnt models are learning and where they fail to learn phonotactic generalizations.

In other words, we compare phonotactic MaxEnt learning models with different constraint sets to find how the presence of particular unnatural constraints affects the model’s success in capturing the data (see also literature like Hayes et al. 2012). We also consider a MaxEnt model that includes faithfulness, learned based on the weights of the model without faithfulness, which is able to produce an even better fit for the data by accounting for the fact that each word has a uniform pronunciation, rather than a probabilistic distribution over pronunciations.

4.1 Setup of MaxEnt Models

All our MaxEnt models are trained on the same training data, which are based on the 1,321 TQ words containing a native morpheme present in the corpus described in Section 3. Tableaux were constructed by starting from the attested surface pronunciation of an input and adding competing surface candidates that explore all alternative

¹²Anttila (2008) alludes to this generalization, but does not provide a more elaborate account. For an earlier treatment of “natural gradient bias”, see Beguš and Nazarov (2018).

possibilities of assigning voicing to the labial and velar stops in the native morphemes in the word. For example, *t̃simpajgu-* ‘to (make) kneel [arodillar(se)]’ will have attested candidate [t̃simpajgu-] and competing candidates [t̃simpajku-, t̃simbajku-, t̃simbajgu-]. An illustrative sample from the training data is given in Section A.4 in the Supplementary Materials. The simulations that will be described in the rest of this section have also been repeated just for the subset of 945 items included in the final model in Section 3.3, which includes only non-derived words with prevocalic instances of labial/velar stops; the results of these are reported in the Supplementary Materials, Section A.9.

Our MaxEnt models assign probabilities to surface candidates that sum to 1 for each underlying representation (UR), as in Goldwater and Johnson (2003) and unlike Hayes and Wilson (2008). We work here with a setup where the underlying voicing of all labial and velar stops is unspecified, e.g., /t̃simPajKu-/ (with P and K being labial and velar stops, respectively, unspecified for [voice]). In this dialect of Quechua, voicing is contrastive, and we do believe that it is ultimately underlying. However, we make this pragmatic modeling decision to simplify learning (see the difficulties with UR learning mentioned by Tesar 2004, 2006, 2014; Jarosz 2006; Nyman and Tesar 2019, amongst others). In Section 4.4, we show that adding voicing contrast to the model while retaining desirable predictions is possible.

In this setup, there are no feature mismatches between URs and surface candidates, which means that IDENT constraints like IDENT(voice) would have no violations for any candidate. Since we do not include any unfaithful mappings except voicing changes in labial/velar stops, meaning that MAX, DEP, and other faithfulness constraints would also yield zero violations, we only include markedness constraints in our simulations, but this is only provisional: if a broader candidate set were considered (one with deletions, insertions, and other unfaithful mappings), various faithfulness constraints would be necessary, so faithfulness must be included in a fuller analysis of the dataset. Each resulting grammar is still compatible with the Rich Base (Prince and Smolensky 2004), since underlying /p/, /b/, /k/, /g/ etc. would still surface as a voiced or voiceless stop, depending on phonotactic considerations.¹³

We consider 42 constraints: 21 typologically and phonetically natural ones and 21 unnatural mirror images. These can be split up into 7 families based on the context in which they operate (Table 7). The constraints whose name starts with * assign one violation mark for every occurrence of the sequence of sounds in the constraint name (as in Section 3, T stands for a voiceless obstruent, D stands for a voiced obstruent, N stands for a nasal, and R stands for a nonnasal sonorant consonant); definitions of (Dis)AGREE and CODACOND will be given below.

The natural constraint families can be justified as follows. There is a general bias against voiced obstruents both across and within languages (Ohala 1983) — hence the constraint *[+voice,-son] or, in short, *D (Lombardi 2001). Voicing (including voiced obstruents) is also generally avoided word-initially (Westbury and Keating 1986), given processes like word-initial strengthening; hence the constraint *#[+voice,-son]. Voicing agreement within clusters is a widely attested phenomenon (Myers 2010); we use the

¹³Surface candidates with unspecified voicing are not included: it is assumed that a high-weighted constraint HAVEVOICE leads to very low probabilities on such candidates.

Family	Natural constraints	Unnatural constraints
Context-free	*[+voice,-son], *b, *g	*[-voice,-son], *p, *k
Word-initial	*#[+voice,-son], *#b, *#g	*#[-voice,-son], *#p, *#k
Voicing agreement	AGREE(voice)/[-son], AGREE(voice)/[+labial,-son], AGREE(voice)/[+dorsal,-son]	DISAGREE(voice)/[-son], DISAGREE(voice)/[+labial,-son], DISAGREE(voice)/[+dorsal,-son]
Coda condition	CODACOND([+voice])/[-son], CODACOND([+voice])/[+labial,-son], CODACOND([+voice])/[+dorsal,-son] ¹³	CODACOND([-voice])/[-son], CODACOND([-voice])/[+labial,-son], CODACOND([-voice])/[+dorsal,-son]
Intervocalic	*VTV, *VpV, *VkV	*VDV, *VbV, *VGv
post-nasal	*NT, *Np, *Nk	*ND, *Nb, *Ng
Post-R	*RT, *Rp, *Rk	*RD, *Rb, *Rg

Table 7 Constraints considered in our MaxEnt models.

AGREE constraint family (Baković 2000).¹⁴ AGREE([F]) assigns one violation mark for every pair of adjacent segments that disagree on [F]. Alternative proposals could be used, instead, e.g., Agreement by Correspondence (Rose and Walker 2004) or Share (McCarthy 2011).

To account for the fact that marked feature values are preferred in onset position or in a coda assimilated to an onset, Itô (1986) proposed the Coda Condition, later adapted as the CODACOND constraint family (e.g., Itô and Mester 1994; Zoll 1998; McCarthy 2007, 2008). CODACOND([F])/[G] assigns one violation mark for every instance of [F] in a coda segment that bears feature [G] such that [F] is not shared with the following onset; e.g., CODACOND([+voice])/[-son] assigns one violation mark for every instance of [-sonorant] in a coda (i.e., a coda obstruent) that is voiced itself but is not followed by a voiced onset.

Finally, constraints against intervocalic (Westbury and Keating 1986; Kaplan 2010; Davidson 2016) and post-nasal (Pater 1999) voiceless obstruents have been proposed to account for the avoidance of such configurations across languages. The *RT constraint family is an extension of these constraints, based on the general idea that a sonorant prefers to be followed by a voiced rather than voiceless obstruent.

The place-specific versions of these constraints (specific to [+labial] or [+dorsal] consonants) are violated whenever the target obstruent in the constraint definition is labial or velar, respectively. The place-specific CODACOND constraints target codas that are labial or velar.¹⁵ The place-specific AGREE constraints are violated whenever at least one obstruent in a cluster is labial (or velar, respectively) and the obstruents disagree in voicing.

The unnatural counterparts of these constraints are obtained by reversing the value of [voice] in the constraint, if said constraint only refers to one of the values (+ or -).

¹⁴We have defined the (DIS)AGREE constraints over obstruents only. One of the reviewers remarked that defining this family over sonorants and obstruents together would allow DISAGREE to trigger post-nasal devoicing as well as post-obstruent voicing. Since post-nasal devoicing is already motivated by *[+voice,-son], as defined above, and the unnatural *ND constraint, to be defined below, we have not pursued this option.

¹⁵The definition of these place-specific CODACOND-constraints is: for every labial obstruent coda (or dorsal obstruent coda, respectively), if it has the feature [+voice], this [+voice] feature is also linked to a segment in onset position.

As the unnatural counterpart of AGREE, we devised the constraint DISAGREE, which assigns a violation mark every time that two adjacent obstruents have equal values of [voice]. Its place-specific variants are defined the same as for the AGREE family: for instance, DISAGREE(voice)/[+labial,-son] is violated once for every obstruent cluster that contains a labial obstruent and in which the obstruents agree in voicing.¹⁶

With these constraints, we built 11 models in total. These comprise 2 basic models: Full (all constraints), Natural (only natural constraints); 8 expanded models, which have a tailored mix of natural and unnatural constraints; and 1 model with faithfulness constraints, which is considered in Section 4.4.

4.1.1 Description of models

The Full model has all 42 constraints in Table 7: the learner weights both natural and unnatural constraints to account for the pattern. This model gives the learner the greatest chance of succeeding and is tested to see what roles the learner assigns to natural vs. unnatural constraints.

The Natural model has all 21 natural constraints and provides a baseline of how well a model with only universal, natural constraints fares on the TQ data.

For each of the 7 unnatural constraint series in Table 7, we consider an expanded model that adds the unnatural constraints from one family to the Natural model. For instance, the *ND model contains the unnatural constraints from the post-nasal family in addition to all natural constraints. Each of these models tests whether including the unnatural constraints from one of the families improves the fit of the resulting optimized model to the data compared to the Natural model, and whether the addition of this constraint is warranted is checked with the likelihood ratio test (which is applicable here since the Natural model is nested inside all the expanded models).

Given the results in Section 3.3, we predict that the addition of unnatural constraints from the post-nasal and Voicing agreement families (*ND, DISAGREE) will lead to an improved fit to the data. If true, this would mean that the gradient phonotactic pattern cannot be modeled without these unnatural constraints. This is what we mean by a “natural gradient bias”: if only natural constraints are present, the unnatural gradient in our data cannot be accounted for.

Based on the latter prediction, we also trained one last model: the DISAGREE & *ND model, which contains unnatural constraints from the Voicing agreement and post-nasal families in addition to the constraints of the Natural model. This model tests whether adding constraints in favor of voicing disagreement and post-nasal devoicing constitutes an improvement over a model that only has voicing agreement and post-nasal voicing constraints.

4.2 Learning setup

All models are trained with Staubs’ (2011) MaxEnt training package, which optimizes constraint weights given an array of tableaux with pre-defined OT-style constraints.

¹⁶For an alternative solution, DISAGREE(voice)/[-son] could be replaced by the combination of OCP(voice) (see Itô and Mester 1986, 2003) and a constraint against two consecutive obstruents sharing the same autosegmental voicing specification. However, we chose the one-constraint option so that natural and unnatural constraints would maximally match each other.

Context	Data points		
Word-initial	/Pata/	→	[pata, bata]
	/Kata/	→	[kata, gata]
Post(/pre)-obstruent	/atPa/	→	[atpa, atba]
	/atKa/	→	[atka, atga]
	/aKPa/	→	[akpa, agpa, akba, agba]
Intervocalic	/aPa/	→	[apa, aba]
	/aKa/	→	[aka, aga]
post-nasal	/amPa/	→	[ampa, amba]
	/aŋKa/	→	[aŋka, aŋga]
Post-R	/alPa/	→	[alpa, alba]
	/alKa/	→	[alka, alga]

Table 8 Test dataset.

Since the data are categorical, the objective of the learner is to maximize data log-likelihood (or, in practice, minimize negative data log-likelihood). The models are optimized with the built-in *optim* function in R (R Core Team 2018) with the L-BFGS-B method (Byrd et al. 1995), L2 optimization with $\mu=0$ and $\sigma=1,000,000$, and a maximum number of iterations of 100,000. Since the models have no hidden structure (cf. Tesar and Smolensky 2000), there is no reason to assume that different runs and different initializations of this algorithm will yield different optima. Nevertheless, each model is run 5 times with all constraint weights initialized at 0. No differences were observed among the 5 runs for any of the models, so the results presented in the next subsection will not refer to distinct runs of the same model.

4.3 Results

The MaxEnt grammars optimized by this method are evaluated quantitatively in terms of the (conditional) log-likelihood on the training data (using the likelihood ratio test and related techniques, which also take the number of constraints into account), and qualitatively in terms of their performance on a test dataset and in terms of their constraint weights. The test dataset consists of labial and velar stops in all relevant environments, as shown in Table 8. Like in the training dataset, surface candidates differ in the voicing of labial and/or velar stops, which allows us to gauge whether the model prefers a voiced or voiceless stop in the specified context.

The purpose of the test dataset is to see how accurately the statistical tendencies in the training dataset are predicted by the grammar (since the training data do not show variation within the same word, we cannot directly predict individual data points). We have no available data on generalization to new items, but, lacking any evidence to the contrary, we assume that matching the frequencies in the lexicon is desirable (e.g., Zuraw 2000; Ernestus and Baayen 2003; Hayes et al. 2009; Moore-Cantwell 2016; Zymet 2018). Obtaining actual generalization data to test this assumption would be an important next step.

Model	Number of parameters	Log-likelihood	AIC	BIC
DISAGREE	24	-609***	1266	1391
DISAGREE/*ND	27	-609***	1272	1412
Full	42	-609***	1302	1520
*ND	24	-714***	1476	1601
Natural	21	-736	1514	1623
*[-voice,-son]	24	-736	1520	1644
*#[-voice,-son]	24	-736	1520	1644
CODACOND([-voice])	24	-736	1520	1644
*VDV	24	-736	1520	1644
*RD	24	-736	1520	1644

Table 9 Results for basic and expanded models.

4.3.1 Results for basic and expanded models

Table 9 shows the results for all basic and expanded models, including the number of param(eter)s (constraints to weight), the log-likelihood ($\log\lambda$) of the training data for the models (closer to zero means a better fit to the data), and the Akaike Information Criterion (AIC, Akaike 1974) and Bayesian Information Criterion (BIC, Schwarz 1978), two measures that score models based on log-likelihood and number of parameters; the lower the AIC/BIC value, the closer the model is to a perfect fit to the training data while being able to generalize. *** indicates a significant difference ($p < .001$) between a model and the Natural model according to the likelihood ratio test (LRT). This test, performed according to NIST (2012), chapter 8.2.3.3, is applied to a pair of models in a superset-subset relationship and asks whether the superset model’s log-likelihood significantly improves upon that of the subset model. We take the difference in number of constraints between the models as the number of degrees of freedom. As can be seen, the Full model and the models that add either DISAGREE or *ND constraints to the Natural model improve on the Natural model on both the LRT and on the AIC/BIC criterion. The weights of the Natural model and these four improving models are given in Table A3 in the Supplementary Materials, section A.5.

To gauge the necessity of adding the DISAGREE and/or *ND constraint families to a MaxEnt model of the TQ data, we compare the DISAGREE and *ND models to the combined DISAGREE/*ND model. The DISAGREE/*ND model constitutes an improvement over the *ND model ($\chi^2(3)=210.04$, $p < .001$), but not over the DISAGREE model ($\chi^2(3)=0$, $p > .5$).¹⁷ Thus, adding the DISAGREE family to a model that already has the *ND family leads to a significant improvement, whereas adding the *ND family to a model that already has the DISAGREE family does not. This means that the *ND family is not necessary: it helps to account for the data compared to the Natural model, but it does not add anything to a model that already has the DISAGREE family.

¹⁷The DISAGREE/*ND model actually performs marginally worse than the DISAGREE model, which would yield a negative χ^2 statistic, but since the latter is impossible, we report 0. The worse performance of the DISAGREE/*ND model most likely stems from the regularization term: there are more constraints in this model, and the regularization term minimizes constraint weights stronger when there are more constraints, meaning that the actual constraint weights are slightly farther removed from the true optimal constraint weights.

On the other hand, the DISAGREE family is necessary, as its presence significantly improves a model that already has the *ND family.

Table 9 shows that the DISAGREE model has the same log-likelihood as the full model when rounded to whole numbers, and, indeed, the addition of all the other unnatural constraint families in the full model does not yield a significant improvement ($\chi^2(18)=0.09$, $p>.5$). This means that the DISAGREE family is, in fact, sufficient to capture the improvement that unnatural constraints can bring to fitting the TQ data.¹⁸

These findings are corroborated by inspecting test data performance. Table 10 shows how well the predictions of the best four models on the test dataset match the training dataset using four of the tendencies in the TQ data; the Natural model is included for comparison. Values that differ from the training data column by ≥ 0.05 are shaded. The attested proportions in the training data are calculated as the proportion of words that start with [p,k] out of all words that start with a labial or velar stop, the proportion of occurrences of a post-obstruent [b,g] out of all obstruent+[p,k,b,g] clusters, the proportion of [mp,ŋk] clusters out of all nasal+[p,k,b,g] clusters, and the proportion of all [Rb,Rg] clusters out of all R+[p,k,b,g] clusters, respectively. The proportions in the test data are calculated as follows: given a context (e.g., word-initial), for the two words for that context (e.g., /Pata/, /Kata/) the model-assigned probabilities of the candidates that obey the tendency active in that context (e.g., [pata], [kata] obey word-initial devoicing) are averaged.

The DISAGREE and DISAGREE/*ND models produce results that are practically identical to the full model. As can be seen in the shaded cells, the Natural and *ND models fail to capture the prevalence of voicing-disagreeing clusters, which are predicted to occur less than 50% of the time, while the Natural model predicts a much lower rate of post-nasal devoicing than observed in the lexicon.

Context	Trend-obeying forms	Attested	Natural	DisAgree	*ND	DisAgree/*ND	Full model
Word-initial	[pata, kata]	0.98	0.98	0.98	0.98	0.98	0.98
Post-obstruent	[atba, atga, a{k,g}ba]	0.80	0.35	0.83	0.41	0.83	0.83
Post-nasal	[ampa, aŋka]	0.90	0.67	0.91	0.91	0.95	0.91
Post-R	[alba, alga]	0.73	0.74	0.74	0.74	0.74	0.74

Table 10 Rate at which four phonotactic tendencies are predicted to be obeyed in relevant forms in the test dataset, according to the best four models.

The behavior of the Natural and *ND models can be interpreted as follows. Since the Natural and *ND models do not have access to the DISAGREE family of constraints, which prefers voicing mismatch in obstruent clusters, the preference for voicing in the second members of clusters cannot be explained by any constraint. This means that there is unexplained voicing variability in the data (voicing after obstruent,

¹⁸At the suggestion of one of the reviewers, we applied stepwise backwards model selection (e.g., Hayes et al. 2012) with significance level $p < .1$ to find, in an unbiased way, a smallest set of constraints that can account for the pattern. The results, see Supplementary Materials section A.6, include natural constraints from all families and unnatural constraints from the DISAGREE and CODACOND families. The constraint CODACOND([-voice]/[+dorsal,-son]), as chosen by the model selection procedure, does not show up as necessary in our main simulations, it can be seen as taking the role of AGREE[voice]/[+dorsal,-son] together with DISAGREE[voice]/[+dorsal,-son], providing an incentive for obstruent clusters starting in a dorsal obstruent to remain fully voiceless (which they do in some words).

devoicing otherwise), which causes the *[+voice,-son] family to receive lower weights (see Table A3 in the Supplementary Materials, section A.5) and not being able to capture post-nasal devoicing. This means that, in the *ND model, the post-nasal devoicing pattern is captured by high-weighted *ND and *Nb, instead, while the Natural model can explain post-nasal voicing to a lesser extent by using the lower-weighted *[+voice,-son] family.

4.4 Accounting for voicing stability within lexical items

4.4.1 Morphophonological learning setup

One important limitation of the models discussed above is that they do not predict categorical voicing in the attested lexical items of TQ, but rather, predict that most labial and velar stops will vacillate between [p,b] and [k,g], respectively, while this is not attested in the dataset (a problem discussed extensively by Zuraw (2010)). For instance, *t̃fimba*- ‘to put a bag on someone’s back [colocar un bulto en la espalda de una persona]’ is only attested with [b] and *t̃fimpa* ‘opposite side (of a watercourse) [orilla contraria]’ is only attested with [p].

This can be remedied by learning URs with a fixed voicing value and including IDENT, so that words with categorical [b] have underlying /b/, words with categorical [p] have underlying /p/, etc. However, the process of learning URs is a complex one (see, e.g., Tesar 2004, 2006, 2014; Jarosz 2006; Nyman and Tesar 2019). Here, we use a two-phase model (cf. Jarosz 2006), in which learning a purely phonotactic grammar, as described in the preceding subsections, is followed by a learning stage that involves specific URs.

For the second stage, we assume that URs are equal to the attested forms in all cases, which creates an incentive for the learner to promote faithfulness constraints over markedness constraints, leading to a lack of generalization (Prince and Tesar 2004). By starting the optimization on the previously learned weights of the best faithfulness-free model (the DISAGREE model), the markedness constraints get a head start. By adding a bias that favors higher markedness constraint weights in relation to the weight of IDENT, the learner is incentivized to make structured predictions about new forms. This markedness-over-faithfulness bias (Prince and Tesar 2004) is implemented by penalizing, during the optimization process, the weight of IDENT minus the combined weights of the markedness constraints, a standard option in the hgR software (Staubs 2011); the degree to which this is done is determined by a coefficient we will call M/F.

4.4.2 Results with faithfulness

The model described above, which may be called DISAGREE/Faithfulness, is trained and tested in the same way as the other models. The *optimal* (best generalizing) version of this model is obtained by setting the initial weights of the markedness constraints to the weights learned by the DISAGREE model, while the initial weight of IDENT is set to 0; and setting the M/F coefficient (controlling the strength of the

markedness-over-faithfulness bias) to 0.0001.¹⁹ This model yields a log-likelihood of about -0.0001 for the entire dataset, which is a nearly perfect performance.

A *baseline* version of the DisAgree/Faithfulness model is also run with initial weights at 0 and without a markedness-over-faithfulness bias, to explore what happens without the additional assumptions of two-stage learning and the markedness-of-faithfulness bias. On the training data, this model also has a near-perfect performance (with a log-likelihood of -0.00003), but it differs in its performance on the test data, as indicated below.

The test dataset is kept identical to the one for the previous simulations, including URs in which the voicing of labial and velar plosives is unspecified, so that IDENT has no effect. This is done to maximize the chance that the model with faithfulness would yield a match with the pattern in the lexicon (and to ensure that the model with faithfulness was tested in the exact same way as the models without faithfulness).

As seen in Table 11, the DISAGREE/Faithfulness model run with optimal parameters yields a close match (within a tolerance of 0.05) to the lexical tendencies for the phonetically natural word-initial devoicing and post-nasal sonorant voicing tendencies, whereas the unnatural post-obstruent voicing and post-nasal devoicing tendencies are still active, but weaker than in the data or the predictions of the DISAGREE model without faithfulness. The DISAGREE/Faithfulness model run with baseline parameters matches the lexical tendencies much less closely than the version with optimal parameters (with the exception of the post-nasal devoicing tendency), but even in this “worst case scenario”, where the markedness constraints are not boosted, each of the tendencies is still observed with a chance above 50%.

The observed discrepancy between the faithfulness-free and faithfulness-included models is to be expected, as the DISAGREE/Faithfulness model does not require markedness constraints to be active to account for the training data, so despite the markedness-over-faithfulness bias in the optimal parameter version, we expect the markedness constraints to be less active in encoding the lexical tendencies (Prince and Tesar 2004). However, this setup does seem to predict that, despite the availability of mechanisms to account for the two unnatural tendencies in the grammar, the unnatural tendencies in this dataset might not match the frequencies in the lexicon as strongly as the natural tendencies. The latter can be tested empirically in future research on TQ.

The reason for the differences between the models is a bit complex. They are not due to attestation in the corpus, since the number of relevant example words is very similar between nasal+/P,K/, sonorant+/P,K/, and obstruent+/P,K/ (125, 149, and 137, respectively). As described in detail in the Supplementary Materials, section A.7, it appears that the lower accuracy on the unnatural tendencies is indirectly related to the unnaturalness of the patterns. If post-nasal devoicing and post-obstruent devoicing had not had natural constraints that favored an attested minority pattern, the accuracy of predicting these tendencies would not have differed so much between the models. In other words, it seems like incongruence between the attested majority pattern and natural constraints leads to less robust learnability. Therefore, there is

¹⁹Higher M/F coefficient values tested did not lead to convergence, while lower values 0.00001 and 0.000001 yielded a decreased performance on the test data. Not using the weights of the initial DISAGREE model or not using a markedness-over-faithfulness bias also yielded degraded test data performance.

also a more general prediction that unnatural tendencies in the lexicon are less robust to learning in at least some conditions.

Of course, this prediction is dependent upon the current setup of the model with faithfulness constraints. In the Supplementary Materials, section A.8, the results of an alternative model with indexed constraints and no faithfulness are summarized. The picture seen in this model is different, but does corroborate the general prediction that unnatural tendencies will be learned less robustly.

Context	Trend-obeying forms	Attested	DisAgree (no faith)	DisAgree/Faith optimal	DisAgree/Faith baseline
Word-initial	[pata, kata]	0.98	0.98	0.93	0.80
Post-obstruent	[atba, atga, a{k,g}ba]	0.80	0.83	0.72	0.54
Post-nasal	[ampa, aŋka]	0.90	0.91	0.70	0.69
Post-R	[alba, alga]	0.73	0.74	0.68	0.54

Table 11 Rate at which four phonotactic tendencies are predicted to be obeyed in relevant forms in the test dataset, according to the DisAgree/Faithfulness model (with the DISAGREE model without faithfulness as a comparison).

These results demonstrate a few interesting points. Firstly, they show the feasibility and learnability of a model that combines the representation of gradient unnatural voicing tendencies (across words) along with lexically fixed voicing values (for individual words); see Moore-Cantwell and Pater (2016), amongst others, for similar work on natural tendencies. Secondly, they show that, even when such models do not match the trends in the lexicon, more strongly misrepresenting the phonetically unnatural trends, these trends are still represented in the grammar at more than 50%, indicating the DISAGREE model’s capability of capturing the data in a generalizable way.

4.5 General discussion of grammar simulations

The Maximum Entropy simulations shown in this section are aimed to test to what extent, if any, grammars that account for the TQ data require unnatural constraints. These simulations have pre-defined natural and unnatural markedness constraints. Based on the literature reviewed in Section 4.1, it is assumed that there are natural constraints against obstruent voicing across the board and word-initially, and in favor of obstruent voicing after nasals, non-nasal sonorants, and intervocally. Furthermore, we assume natural constraints for voicing agreement between adjacent segments (AGREE(voice)/[-son]) and against voicing in codas unless followed by a voiced obstruent (CODACOND([+voice])/[-son]). These constraints are inverted in ways specified in Section 4.1 to yield unnatural counterparts. We compare a model with only natural constraints to various models that also contain unnatural constraints.

The model with unnatural voicing disagreement constraints and only natural constraints otherwise (the DISAGREE model) comes out as the best one: it yields the maximal increase in fit to the training data compared to the Natural model (equivalent to the one achieved by adding all unnatural constraints) with the addition of just one unnatural constraint family. The model that includes both voicing disagreement and post-nasal devoicing constraints performs equivalently, while a model that includes

only post-nasal devoicing constraints performs worse. This implies that an unnatural constraint promoting voicing disagreement is necessary, while an unnatural constraint promoting post-nasal devoicing is not: post-nasal devoicing can be accounted for by giving high weight to the context-free natural constraint $*[+voice, -son]$, and the best possible performance on the training data can be achieved without adding a post-nasal devoicing constraint.

Finally, including faithfulness can ensure that there is no within-word variation while still maintaining probabilistic phonotactic tendencies in the grammar. Our DISAGREE/Faithfulness model accounts for the data near-perfectly, assuming that the URs are equal to the observed surface forms. However, in terms of generalization, this model does not represent the unnatural tendencies as strongly as the best faithfulness-free model. Nevertheless, even with non-optimal learning settings, both natural and unnatural trends in the lexicon are generalized to some extent by this DISAGREE/Faithfulness model.

Thus, while TQ exhibits two statistically significant phonetically unnatural tendencies – post-obstruent voicing and post-nasal devoicing (Section 3.3) – our simulations suggest that only post-obstruent voicing requires unnatural constraints in the grammar. Gradient post-nasal devoicing in our dataset can be modeled as an interaction between a low-weighted constraint in favor of post-nasal voicing ($*NT$) and a high-weighted constraint against obstruent voicing ($*[+voice, -son]$), since voiced obstruents are still very much a minority in the native vocabulary. This means that, against our expectations, the Tarma Quechua phonotactic data do not provide evidence for a post-nasal devoicing constraint in the phonological grammar. If the proportion of voiced obstruents had been higher across the board in this language while the proportion of voiced obstruents after nasals had remained the same, there might have been phonotactic evidence for a phonologically unnatural constraint $*ND$. While TQ phonotactics do not support the $*ND$ constraint, the *morphophonological* alternations shown in Section 3.5 that are limited to a subset of morphemes might provide independent evidence for $*ND$. Testing the productivity of these alternations and their theoretical implications is left for future work. Additionally, there are other languages that provide strong evidence in favor of the $*ND$ constraint: Hyman 2001; Coetzee and Pretorius 2010; Beguš 2019.

The post-obstruent voicing pattern, however, does require an “unnatural” constraint in the grammar, DISAGREE(voice)/[-son], which penalizes adjacent obstruents that agree in voicing. According to the definition in Section 1.2, an unnatural pattern is one that runs counter to a universal phonetic tendency. By extension, a constraint that consistently favors such unnatural patterns, in this case, voicing-disagreeing obstruent clusters, is seen as an unnatural constraint.

One uniquely unnatural aspect of the effects of DISAGREE(voice)/[-son] in TQ is that it targets two adjacent obstruents. Phonetic pressures towards voicing uniformity in these conditions operate as strong phonetic tendencies across languages (see discussion in Section 1.2). While voicing dissimilation is attested (cf. Dahl’s Law in Gikuyu, Lombardi 1995 and references cited therein, and Rendaku in Japanese, e.g., Itô and Mester 2003), it typically operates non-locally, where the articulatory phonetic tendencies towards voicing agreement are less strong. Another uniquely unnatural aspect

arises from the interaction of DISAGREE(voice)/[-son] with the (natural) CODA-COND([+voice])/[-son] constraint, namely, a preference for output clusters that contain voicing in the second member (TD) rather than in the first member (DT; see Table 6 and Section 3.3.2 for why this distribution is phonetically unnatural). As far as we can see, DISAGREE(voice)/[-son] does not drive any phonetically natural patterns in TQ in addition to the phonetically unnatural voicing disagreement pattern, meaning that this is truly an unnatural constraint according to our definition.

Despite its clear phonetic unnaturalness, a constraint like DISAGREE may have a functional motivation, as it promotes dissimilation. For instance, Bennett (2015) provides an account of rhotic dissimilation in Sundanese motivated by the combination of pressures to have surface rhotics correspond to one another (Rose and Walker 2004) and to limit such correspondences when the two rhotics are not onsets of adjacent syllables. However, it can be argued that such pressures are rather of a structural or cognitive character, not contradicting the phonetic unnaturalness of the process. Moreover, TQ exhibits segmentally local dissimilation, which (Bennett 2015, §9.2.2) argues shows different typological patterns from long-distance dissimilation and may have a different motivation.

We do not take a specific stance on the origin of unnatural constraints here. In the literature, there is doubt about the productivity of (some) unnatural patterns (e.g., Hayes et al. 2009; Becker et al. 2011, 2012). If unnatural constraints are indeed learned, not universal, then the question of constraint induction comes into play (e.g., Hayes 1999; Hayes and Wilson 2008). Unfortunately, a full investigation of this topic is outside the scope of this paper, but the current simulations do show that adding DISAGREE constraints to a model with natural constraints significantly increases the fit to the training data, which shows that the existing data provide evidence for a learner to induce (a) constraint(s) in favor of post-obstruent voicing.

Finally, despite the fact that it has been shown that Maximum Entropy models are not restricted to generating phonetically natural patterns when the grammar only contains natural constraints (see, e.g., Magri and Anttila 2018), the results of the current simulations show that there is still a “natural gradient bias” in these types of models: not all (gradient) phonetically unnatural patterns can be modeled by interactions of natural constraints alone, and the presence of certain unnatural constraints leads to a significantly better fit to the training data when modeling datasets such as the TQ one. See also Hayes (2016) and O’Hara (2022) for similar results showing that an all-natural constraint set imposes restrictions on possible (gradient) patterns in Maximum Entropy models.

If it turns out that the unnatural patterns in the TQ lexicon (particularly post-obstruent voicing) do not generalize in any way to novel items, the motivation for the inclusion of DISAGREE becomes much weaker. However, the simulations still have made it clear that at least some productivity of the post-nasal devoicing tendency is expected, and a lack of productivity should be accounted for by a separate mechanism.

5 Conclusion

This paper argues that phonological grammar can be probabilistic and operate in phonetically unnatural directions at the same time. We present a case of unnatural gradient phonotactic restrictions in the lexicon: Tarma Quechua stop voicing. We show that, in the data, the phonotactic restrictions operate in an unnatural direction (as defined in Section 1.2): in the environments presented (post-nasal, post-voiceless-obstruent, and intervocalic), the articulatorily or perceptually preferred value of $[\pm\text{voice}]$ is less frequent than the dispreferred value. Subsequent analysis (see below) shows that the post-voiceless-obstruent environment requires unnatural elements in the phonological grammar, while the other unnaturalness might be an epiphenomenal consequence of a general bias against voiced obstruents in the language. Based on a new corpus (which includes both published and unpublished data) we argue that the unnatural phonotactic restriction is statistically significant and phonetically real. This appears to be the first report of unnatural gradient phonotactics that target segmental features in a segmental context. In Section 3.5, we provide evidence for the productivity of these phonotactic restrictions, which suggests that they are actually present in the grammar (see Coetzee and Pater 2008 and others for arguments that gradient phonotactic patterns must be present in the synchronic grammar).

We furthermore use grammar learning simulations to test whether some phonetically unnatural gradient phonotactic restrictions can be derived as a by-product of natural constraint interactions without any unnatural constraints. We argue that at least some phonetically unnatural constraints are necessary in the grammar to model the Tarma Quechua data to their full extent. Interestingly, our results (with and without faithfulness constraints) provide evidence for a post-obstruent voicing constraint, but not for a post-nasal devoicing constraint. This is because post-nasal devoicing can be parasitically accounted for by a general devoicing constraint, given that the data show a general dispreference for voiced obstruents, while post-obstruent voicing goes against the general devoicing trend in the data.

This result is obtained by evaluating the necessity of unnatural constraints using the goodness of fit of individual models. This approach allows us to evaluate analyses with natural and unnatural constraints or models with and without indexed constraints. The results of this approach can serve as a diagnostic of which constraints are needed for a model that provides the best fit to the data. The proposed approach also helps us interpret how the MaxEnt learners derive the data. For example, it has helped us understand how post-nasal devoicing can be derived without unnatural constraints. The proposed approach can be used both in further discussions on natural and unnatural constraints as well as for evaluating other phonological analyses.

In future work, we hope that our evidence for the gradient unnatural phonotactic patterns presented here will be corroborated with experimental data. Because of the relative inaccessibility of the Tarma speech community, obtaining such experimental data was outside the scope of this research. However, this type of evidence would greatly enhance the strength of our claim that phonological grammars should at the

very least allow some phonetically unnatural constraints.²⁰ In particular, one prediction of the current analysis is that unnatural tendencies in the lexicon might be less robustly learned compared to natural tendencies. Another related direction for future work would be to test the productivity of the limited morphophonological alternations mentioned in Section 3.5 and computationally model these patterns, to see whether they might provide evidence for the activity of a post-nasal devoicing constraint (*ND) in the grammar.

Furthermore, we hope this paper will inspire exploration of other cases of gradient unnatural phonotactics, particularly ones that involve a feature other than [$\pm$ voice]. Finding phonotactic restrictions that gradiently favor the phonetically unnatural variant is a challenging task which has thus far not been a focal area within phonological theory. This paper draws attention to the importance of such processes, points to their theoretical implications, and provides a quantitative framework for evaluating and interpreting weighted-constraint approaches to unnatural phonotactics. We hope this work will spark further interest in phonological processes that are simultaneously probabilistic and unnatural, and we expect that more such cases will be discovered in the future.

Data Availability

Code and data are available at:
<https://doi.org/10.17605/OSF.IO/9Y2XG>.

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Competing interests

The authors declare no competing interests.

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²⁰Whether unnatural constraints are universal/innate or learned from the data is a separate question, one that we do not take a stance on in this paper.

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Appendix A Supplementary Materials

A.1 Data collection and annotation

Each target stop’s syllabic association was annotated right-to-left, with the first counted syllable therefore being the rightmost (thus backwards to English reading order). Syllabic association was furthermore annotated in two ways: firstly via syllabifying each word according to normal Tarma Quechua syllable structure, and secondly via syllabifying each word in order to maximize onsets, regardless of whether these are actually licensed in the language. The latter technique is used out of an abundance of caution, in order to simplify the process of analyzing target CC clusters. Tarma Quechua does not typically tolerate within-syllable CC clusters, although a stop followed by a glide or liquid seems to be permitted, at least in loans. This can be seen in examples such as *kwarta* [Sp. *cuarto*] ‘room’ and *gra:tis* [Sp. *gratis*] ‘thanks’. Because of examples such as these, consonant + liquid or consonant + glide clusters were kept when syllabifying according to the actual phonotactics of the language. Consonant + nasal clusters were not syllabified together, as evidence is marginal that these are permitted. The data also contained several glide + vowel or vowel + glide sequences; these sequences were treated as one vowel V.

Part of speech was also annotated in two ways. The first annotation simply recorded the tagging of each source text; however, each author’s tagging system did not exactly align. In order to provide consistency across part-of-speech tagging, we collapsed the tagging systems of both authors into a simple four-way system of nouns, verbs, dual noun/verbs, and “other” parts of speech (e.g. adjectives, adverbs, interjections, etc). Closed-class nouns (e.g. pronouns, numbers) were not tagged as nouns for the purpose of this analysis; instead, they were tagged as “other.”

Duplicate entries between Adelaar’s and Baldoceda’s dictionaries were treated in several ways. If multiple identical-looking entries appeared together in the same dictionary entry, they were treated as unique and not consolidated together, as in the case of homophones. If a word was entered multiple times under the same entry in a single dictionary, all were considered a single entry (e.g. a single word with both verbal and adjectival senses, listed separately, would be marked as one entry). If two entries occurred in separate dictionaries, but one appeared to be a hypernym of the other, the two were consolidated into one entry (e.g. *parba* ‘food offering’, and *parba* ‘sweet corn’ [‘dulce de maiz’]). Lastly, entries for which one definition seemed like a plausible semantic extension of the other were still marked as separate entries, in the absence of explicit confirmation of their relationship.

Because alveolars never undergo voicing, they were omitted from the analysis; only labials and velars were kept. In addition, word-final stops and the first members of consonant clusters almost always surface as voiceless in native vocabulary, which is why they were also excluded from the analysis. For the purpose of statistical analysis, we only test consonants in prevocalic position. Morpheme boundary and word-boundary are collapsed into one category (lexical boundary, marked with #). To test significance of the observed trends in the lexicon, we include only native vocabulary in the analysis and exclude words marked as derived in order to avoid overrepresenting patterns in those lexical items that are frequently derived ($N = 1,118$). We also perform a

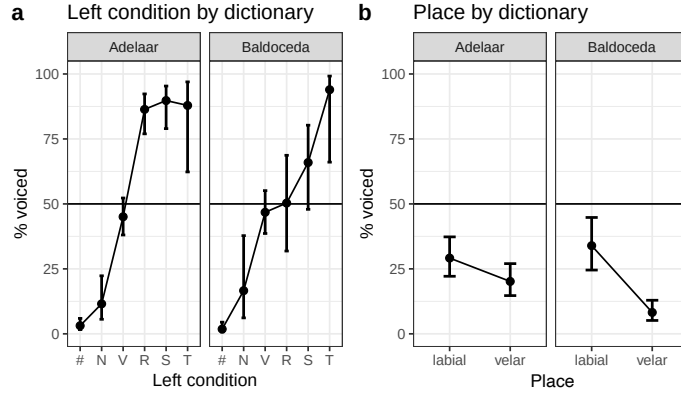


Fig. A1 Estimates of a logistic regression model with PLACE, LEFT CONDITION, and DICTIONARY as predictors and two interactions: LEFTCOND:DICTIONARY and PLACE:DICTIONARY (non-loan non-derived vocabulary only).

significance test on vocabulary that includes loanwords and derived items to test whether observed patterns persist when the entire vocabulary is included.

A.2 Models

The initial full model was fit with all predictors and their interactions (AIC = 930.01). The best fitting model was chosen with the step-wise backwards model selection technique: higher order interactions were removed step-wise from a full model. If the Akaike Information Criterion (AIC) is decreased by removing an interaction or predictor, they were removed until all predictors in the model improved the fit. The best-fitting model includes PLACE, LEFT CONDITION, and DICTIONARY as predictors and two interactions: LEFTCOND:DICTIONARY and PLACE:DICTIONARY (AIC = 876.29). The PART OF SPEECH predictor does not substantially improve model fit according to the AIC test.

The DICTIONARY predictor has two levels: Adelaar includes all data entries that are in Adelaar or in both dictionaries; Baldoceda includes those entries that appear in the Baldoceda dictionary only and are absent from Adelaar. This predictor thus indicates whether either data source deviates significantly from the other. Figure A1 in Section A.2 in the Supplementary Materials shows that the major trends in Adelaar's dictionary hold in those lexical items in Baldoceda that are absent from Adelaar. This suggests that the voicing pattern is independently confirmed in two sources. However, because the predictor DICTIONARY is not linguistically justifiable (lexical items that are absent from Adelaar 1977, but present in Baldoceda have no linguistic function; their only function is to test whether the trends persist across dictionaries), we exclude this predictor from the final model.

A.3 Analysis of reverberation

Due to generally sub-optimal recording conditions, reverberation is present throughout the audio analyzed here. However, we can still reliably distinguish between

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)=V	-0.1623	0.0852	-1.91	0.0567
#	-2.0487	0.1671	-12.26	0.0001
N	-1.7818	0.3061	-5.82	0.0001
R	1.2640	0.2147	5.89	0.0001
S	1.1647	0.2368	4.92	0.0001
T	2.3277	0.6120	3.80	0.0001
Place1	0.5264	0.0852	6.18	0.0001
#:Place1	0.2004	0.1671	1.20	0.2302
N:Place1	-0.8849	0.3061	-2.89	0.0038
R:Place1	0.2589	0.2147	1.21	0.2278
S:Place1	0.0582	0.2368	0.25	0.8059
T:Place1	-0.8460	0.6120	-1.38	0.1669

Table A1 Estimates of a logistic regression model with PLACE (sum-coded) and LEFT CONDITION and their interaction as predictors with loans and derived words included in the analysis.

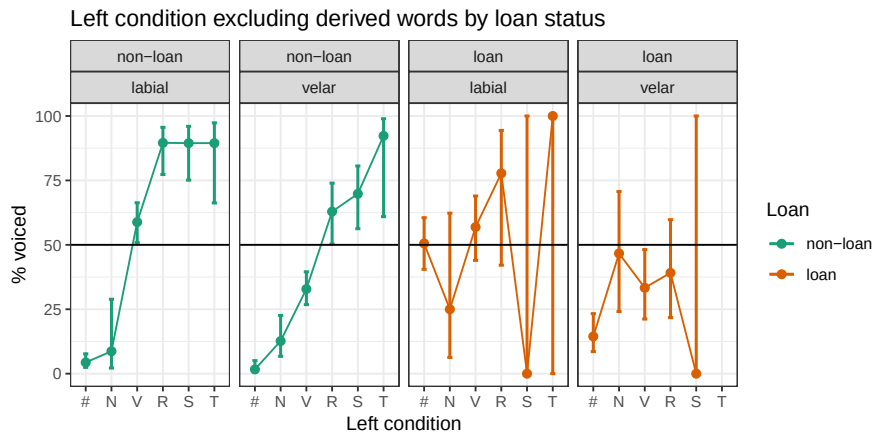


Fig. A2 Estimates of a logistic regression model with three predictors, LEFT CONDITION, PLACE, and LOAN with all interactions (derived words excluded).

energy caused by the reverberation of the preceding vowel and voicing. This is done by comparing the suspected reverberation in two environments: word-medially and utterance-finally.

Figure 2 shows four utterances beginning with a VT sequence. The initial vowel is either [a] or [u]. We can see from these examples that following the ending of the initial vowel, energy is still visible in the waveform and spectrogram. This energy seems to last for approximately 75-100 ms, before the signal reverts to reflecting the basic background noise of the room.

If this energy were reverberation, we would expect that it would still be present utterance-finally; that is, when there is no possible following voiced sequence. If this energy were voicing, we would not expect to see it utterance-finally for the same reason. Figure A4 shows the vowels from Figure 2 utterance finally. It is evident from

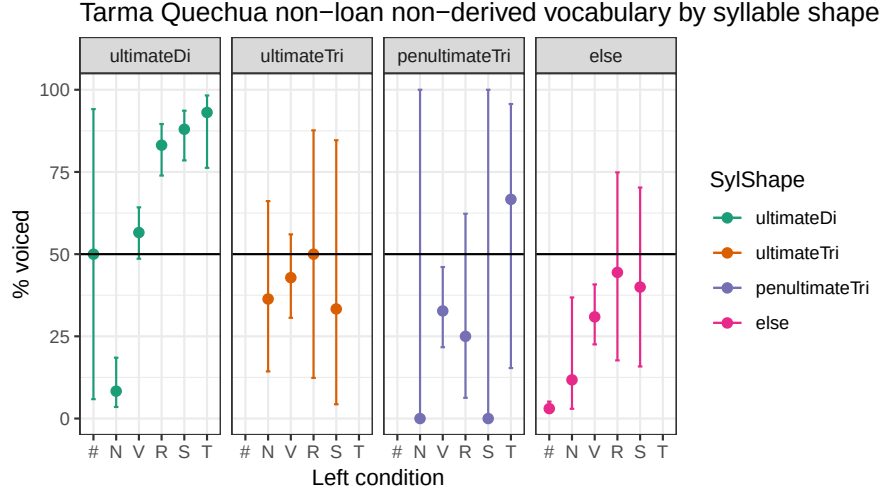


Fig. A3 Estimates of a logistic regression model with LEFT CONDITION and SYLLABLE STRUCTURE as predictors with their interaction.

these utterance-final vowels that some vibration is still present after the end of the vowel, again for around 75-100 ms. This supports the interpretation that this energy in both cases is reverberation, and not voicing.

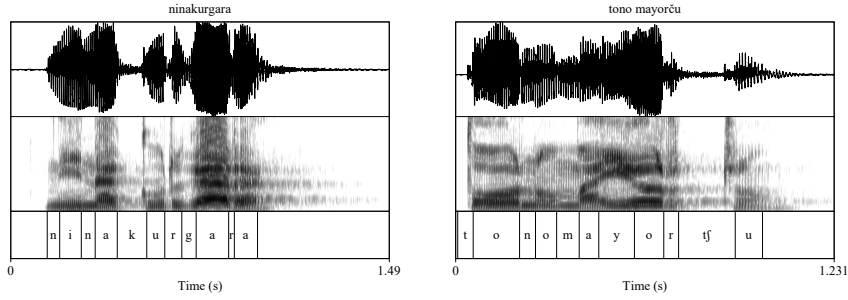


Fig. A4 Waveform and spectrogram of utterance-final [a] and [u], demonstrating reverberation in (left) utterance-final [a] in [ninakurgara], (right) utterance-final [u] in [mayorču].

That the energy present in the initial consonant segment of TD clusters in Figure 2 is not voicing can also be shown through comparing each putative T segment with its putative D segment. The voiced segments each have a noticeably darker bar at the bottom, low-frequency end of their respective spectrograms, indicative of voicing. Clear periodicity can also be seen on their respective waveforms throughout the segment, again indicative of voicing. While the T segments do show some periodicity in the beginning of their waveforms, this is not unexpected, and likely represents the periodic reverberation of the previous vowel during the articulatory transition to a voiceless stop.

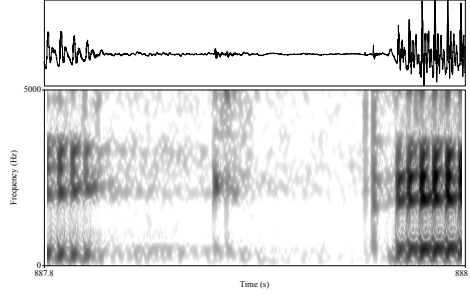


Fig. A5 A waveform and spectrogram of a voiceless cluster [kt] cluster in a relatively silent recording.

UR	Output	Attested	*D	AGREE(vce)/[-son]	CODACOND(+voice)/[-son]	*NT	*VTV
/t̥simPajKu-/	[t̥simpajgu-]	1	-1	0	0	-1	0
/t̥simPajKu-/	[t̥simpajku-]	0	0	0	0	-1	-1
/t̥simPajKu-/	[t̥simbajgu-]	0	-2	0	0	0	0
/t̥simPajKu-/	[t̥simbajku-]	0	-1	0	0	0	-1
/qiPt̥u/	[qipt̥u]	1	0	0	0	0	0
/qiPt̥u/	[qibt̥u]	0	-1	-1	-1	0	0

Table A2 Fragment of the training data. Output candidates explore all possible ways of making underlying /P,K/ voiced or voiceless. 5 of the 21 natural constraints are shown here.

As a comparison, we also add a voiceless in Figure A5.

A.4 Fragment of training data for grammar learning simulations

Table A2 shows a fragment of the training data with a few of the 21 natural constraints. This is to illustrate what the training data look like. Two inputs are included, and the attested candidates are marked with 1.

A.5 Constraint weights in selected models

The table below shows the weights obtained for the Full model, the Natural model, the faithfulness-free DISAGREE/*ND (D.A./*ND), *ND, and DISAGREE (D.A.) models, and the DISAGREE/Faithfulness (D.A./Faith) model.

A.6 Results of stepwise backwards model selection in grammar learning simulations

Applying a stepwise backwards model selection procedure (e.g., Hayes et al. 2012) to the training data with significance level $p < .1$ yields the following model with a training data log-likelihood of -610:

A.7 Correlation between violations of constraints

As can be seen in Table A3 in section A.5, all Markedness constraint families have lower weights in the DISAGREE/Faithfulness model compared to the Faithfulness-free DISAGREE model, except the $*\#[+voice, -son]$, $*NT$, AGREE, and CODACOND([+voice]) families (CODACOND([+voice]) itself has a slightly lower weight, but its two place-specific versions do receive a higher weight). The effect of a higher weight of $*\#[+voice, -son]$ is observed in a better match to the initial devoicing tendencies than predicted by $*[+voice, -son]$ alone, while higher weights for $*NT$, AGREE, and CODACOND([+voice]) lead to worse matches to the post-nasal devoicing and post-obstruent voicing tendencies. Table A5 reveals that $*NT$ is the only constraint family whose violations have a negative pairwise correlation with IDENT (because $*NT$ prefers a minority pattern – post-nasal voicing – and IDENT in this setup always prefers attested candidates), while $*\#[+voice, -son]$, AGREE, and CODACOND([+voice]) are the only constraint families whose violations are positively correlated with those of $*[+voice, -son]$ (because they prefer a voiceless outcome). L2 regularization, used in the current simulations, leads to weights’ being distributed between positively correlated (collinear) constraints (Hoerl and Kennard 1970). Since IDENT has non-zero weight in the model with Faithfulness, it is predicted that the weights of the constraints positively correlated with it would be lowered compared to the ones in the model without Faithfulness (in which IDENT can be seen as having zero weight), leaving out $*NT$, which now has the chance to receive a higher weight. Since the weights of the $*[+voice, -son]$ family are lower in the model with Faithfulness compared to the model without Faithfulness, it is to be expected that the weights of positively correlated constraint families $*\#[+voice, -son]$, AGREE, and CODACOND([+voice]) would be higher.

A.8 Alternative: indexed constraints instead of faithfulness

Instead of the faithfulness constraint IDENT, our indexed constraint model has an indexed version (Pater 2000) of each markedness constraint that has violations for every word in which the constraint is winner-preferring (all natural and unnatural constraints in Table A3 in section A.5 were used here). After this model is trained with the same procedure as our other models, it is tested on the test data without using the indexed constraints. The results are as follows, using different values for the variance in the L2 regularization prior:

The picture painted here is quite different from the model with faithfulness, as post-obstruent voicing is actually overlearned instead of underlearned, and there is a greater discrepancy between predicted and attested probabilities for post-sonorant voicing. However, it looks like post-nasal devoicing is not learned at all, while the predicted rate of post-obstruent voicing has a broad range of variation (0.15), while the other three patterns remain within a range of 0.1 or less. This means that the unnatural tendencies are still less robust than the natural ones, despite a less clear picture than in the faithfulness model presented in the paper.

A.9 Results of simulations with non-derived words only

The simulations as described in section 4 are also applied to the subcorpus of non-derived words with prevocalic labial/velar stops, which contains 945 items. The extent to which these models capture the phonotactic tendencies is shown in Table A7, which shows a very similar pattern to the simulations with the entire corpus presented earlier. In the same way as found for the entire corpus, the Full, DISAGREE, *ND, and DISAGREE/*ND models achieved a significantly higher log-likelihood than the Natural model ($\chi^2(3)=91.76$, $p<.001$), but not over the DISAGREE model ($\chi^2(3)=0$, $p>.5$), see Table A8. As with the larger dataset, the DISAGREE/*ND and DISAGREE models have the same log-likelihood as the full model when rounded to whole numbers, and, indeed, the addition of all the other unnatural constraint families in the full model does not yield a significant improvement over the DISAGREE/*ND model ($\chi^2(15)=0.37$, $p>.5$).

Applying a stepwise backwards model selection procedure (e.g., Hayes et al. 2012) with significance level $p < .1$ to the smaller dataset yields the model shown in Table A9 with a training data log-likelihood of -467. The table shows that a DISAGREE constraint has indeed been selected, in addition to *Nb, even though the *ND family does not appear to be necessary based on the main set of simulations. However, *Nb appears to have the function of ensuring that the rate of post-nasal non-voicing is different between labials and dorsals, which could also have been achieved by including *Np and *Nk instead of *Nb and *NT.

Just as with the larger dataset, simulations with Faithfulness were also run. The optimal Markedness-over-Faithfulness coefficient was 0.0001 (log-likelihood = -0.00007), and a baseline model was also run (log-likelihood = -0.00003). As can be seen in Table A10, all observed tendencies in the lexicon still have a probability of at least 0.5 in the models with Faithfulness.

Table A11 shows the weights obtained for the Full model, the Natural model, the faithfulness-free DISAGREE/*ND (D.A./*ND), *ND, and DISAGREE (D.A.) models, and the DISAGREE/Faithfulness (D.A./Faith) model. As can be seen, the general picture is very similar to the picture for the larger dataset, except that the weights for the markedness constraints in the D.A./Faith model are lower than for the larger dataset. We speculate that this may have to do with the removal of many of the non-prevocalic labial/velar stops from the data, which may have increased the proportion of labial/velar stops whose voicing is predictable by phonotactics, which, in turn, might have increased the collinearity between IDENT and the markedness constraints, making it more difficult for the learner to disentangle their effects.

Constraint	Full	Natural	D.A./*ND	*ND	D.A.	D.A./Faith
Ident						15.996
<i>Natural markedness constraints</i>						
*[+voice,-son]	3.177	0.164	2.446	0	2.451	1.601
*b	1.006	0.129	1.034	0	1.029	0.724
*g	2.171	0.974	1.847	0.816	1.842	1.399
*#[+voice,-son]	1.753	2.076	0	2.280	0	0
*#b	1.128	1.095	0	1.182	0	0.004
*#g	0.625	0.981	0	1.098	0	0.005
AGREE(voice)/[-son]	0.157	0	0.142	0	0.233	0.856
AGREE(voice)/[+labial,-son]	0.442	0	0.555	0	0.688	0.999
AGREE(voice)/[+dorsal,-son]	0.111	0	0.553	0	0.600	1.083
CODACOND([+voice])/[-son]	4.701	2.326	4.331	2.520	4.423	3.847
CODACOND([+voice])/[+labial,-son]	0.001	0	0.093	0.075	0	0.033
CODACOND([+voice])/[+dorsal,-son]	5.700	9.618	6.614	11.044	8.758	8.771
*VTv	2.792	0.300	2.394	0.032	2.394	1.685
*VpV	1.520	0.252	1.344	0.227	1.344	0.837
*VkV	1.417	0.101	1.161	0.046	1.162	1.066
*NT	1.478	0	1.278	0.282	0.711	0.813
*Np	0.237	0	0.325	0.475	0.228	0.431
*Nk	1.376	0	1.275	0.278	1.516	1.510
*RT	3.727	1.243	3.357	1.038	3.357	2.274
*Rp	2.075	0.857	1.930	0.770	1.931	1.296
*Rk	1.651	0.386	1.427	0.268	1.427	1.085
<i>Unnatural markedness constraints</i>						
*[-voice,-son]	0.484					
*p	0.175					
*k	0.309					
*#[-voice,-son]	1.882					
*#p	1.053					
*#k	0.857					
DISAGREE(voice)/[-son]	4.317		4.716		4.803	4.339
DISAGREE(voice)/[+labial,-son]	1.831		1.215		1.351	1.232
DISAGREE(voice)/[+dorsal,-son]	1.702		1.198		1.249	0.960
CODACOND([-voice])/[-son]	0.270					
CODACOND([-voice])/[+labial,-son]	0					
CODACOND([-voice])/[+dorsal,-son]	0.953					
*VDV	0.286					
*VbV	0.243					
*VgV	0.106					
*ND	0.186		0.209	1.685		
*Nb	0.548		0.454	1.559		
*Ng	0.181		0.117	0.126		
*RD	0.111					
*Rb	0.358					
*Rg	0.221					

Table A3 The best performing models and the weights assigned to them by the learner.

Constraint	Weight
<i>Natural markedness constraints</i>	
*[+voice,-son]	3.268
*g	1.206
CODACOND([+voice]/[+labial,-son])	4.393
CODACOND([+voice]/[+dorsal,-son])	15.273
*VpV	3.526
*VkV	3.736
*Nk	2.406
*RT	5.000
<i>Unnatural markedness constraints</i>	
DISAGREE(voice)/[+labial, -son]	5.021
DISAGREE(voice)/[+dorsal, -son]	5.400
CODACOND([-voice]/[+dorsal,-son])	4.757

Table A4 Selected constraints and their weights resulting from applying stepwise backwards model selection.

	corr with Ident	corr with *[+voice,-son]
*[+voice,-son]	0.57	-
*b	0.29	-
*g	0.44	-
*#[+voice,-son]	0.46	0.48
*#b	0.31	0.33
*#g	0.30	0.31
AGREE(voice)/[-son]	0.13	0.33
AGREE(voice)/[+labial,-son]	0.07	0.22
AGREE(voice)/[+dorsal,-son]	0.13	0.25
CODACOND([+voice])/[-son]	0.21	0.18
CODACOND([+voice])/[+labial,-son]	0.08	0.09
CODACOND([+voice])/[+dorsal,-son]	0.19	0.16
*VTV	0.09	-0.21
*VpV	0.13	-0.12
*VkV	0.005	-0.17
*NT	-0.14	-0.17
*Np	-0.08	-0.09
*Nk	-0.11	-0.14
*RT	0.08	-0.17
*Rp	0.10	-0.11
*Rk	0.01	-0.13
DISAGREE(voice)/[-son]	0.08	-0.12
DISAGREE(voice)/[+labial,-son]	0.09	-0.06
DISAGREE(voice)/[+dorsal,-son]	0.05	-0.07

Table A5 Correlations between the violations of markedness constraints and IDENT and *[+voice,-son].

Context	Trend-obeying forms	Attested	$\sigma = 1,000,000$	$\sigma = 10,000$	$\sigma = 1,000$	$\sigma = 600$
Word-initial	[pata, kata]	0.98	0.96	0.91	0.90	0.88
Post-obstruent	[atba, atga, a{k,g}ba]	0.80	1.00	0.97	0.88	0.85
Post-nasal	[ampa, aŋka]	0.90	0.41	0.48	0.50	0.49
Post-R	[alba, alga]	0.73	0.57	0.60	0.64	0.67

Table A6 Rate at which four phonotactic tendencies are predicted according to indexed constraint models with four different values of the regularization prior.

Context	Trend-obeying forms	Attested	Natural	DisAgree	*ND	DisAgree/*ND	Full model
Word-initial	[pata, kata]	0.97	0.97	0.98	0.97	0.98	0.97
Post-obstruent	[atba, atga, a{k,g}ba]	0.81	0.41	0.83	0.42	0.83	0.83
post-nasal	[ampa, aŋka]	0.88	0.58	0.90	0.90	0.90	0.90
Post-R	[alba, alga]	0.75	0.74	0.74	0.74	0.74	0.74

Table A7 Rate at which four phonotactic tendencies are predicted to be obeyed in relevant forms in the test dataset, according to the best four models.

Model	Number of parameters	Log-likelihood	AIC	BIC
DISAGREE	24	-467***	982	1098
DISAGREE/*ND	27	-467***	988	1119
Full	42	-467***	1018	1221
*ND	24	-513***	1074	1190
Natural	21	-533	1109	1211
*[-voice,-son]	24	-531	1109	1226
*#[-voice,-son]	24	-533	1115	1231
CODACOND([-voice])	24	-533	1115	1231
*VDV	24	-533	1114	1230
*RD	24	-533	1115	1231

Table A8 Results for basic and expanded models for non-derived words with prevocalic labial/velar stops.

Constraint	Weight
<i>Natural markedness constraints</i>	
*b	3.306
*g	4.333
CODACOND([+voice]/[+labial,-son])	3.529
CODACOND([+voice]/[+dorsal,-son])	12.696
*VTV	3.577
*NT	2.356
*Rp	5.055
*Rk	4.861
<i>Unnatural markedness constraints</i>	
DISAGREE(voice)/[-son]	5.224
*Nb	1.448

Table A9 Selected constraints and their weights resulting from applying stepwise backwards model selection.

Context	Trend-obeying forms	Attested	DisAgree (no faith)	DisAgree/Faith optimal	DisAgree/Faith baseline
Word-initial	[pata, kata]	0.97	0.98	0.87	0.80
Post-obstruent	[atba, atga, a{k,g}ba]	0.81	0.83	0.55	0.53
Post-nasal	[ampa, aŋka]	0.88	0.90	0.63	0.63
Post-R	[alba, alga]	0.75	0.74	0.58	0.56

Table A10 Rate at which four phonotactic tendencies are predicted to be obeyed in relevant forms in the test dataset, according to the DisAgree/Faithfulness model (with the DISAGREE model without faithfulness as a comparison).

Constraint	Full	Natural	D.A./*ND	*ND	D.A.	D.A./Faith
Ident						16.227
<i>Natural markedness constraints</i>						
*[+voice,-son]	3.347	0	1.893	0	1.985	0.484
*b	0.030	0	0.610	0	0.482	0.135
*g	3.317	0.690	2.209	0.468	2.116	0.482
*#[+voice,-son]	1.078	2.241	0	2.315	0	0.678
*#b	1.692	1.140	0.888	1.066	0.925	0.491
*#g	0.004	1.101	0	1.249	0	0.399
AGREE(voice)/[-son]	0	0	0.284	0	0.073	0.041
AGREE(voice)/[+labial,-son]	0.987	0	0.571	0	1.004	0.102
AGREE(voice)/[+dorsal,-son]	0.004	0	0.435	0	0.561	0.102
CODACOND([+voice])/[-son]	3.328	1.394	3.290	1.106	3.167	0.452
CODACOND([+voice])/[+labial,-son]	0	0	0.023	0.293	0.141	0.279
CODACOND([+voice])/[+dorsal,-son]	5.473	10.581	5.956	11.406	12.965	0.385
*VTV	3.042	0	1.904	0	1.956	0.516
*VpV	0.337	0.288	0.886	0.288	0.798	0.268
*VkV	2.860	0	1.428	0	1.375	0.161
*NT	1.963	0	0.888	0.288	0.032	0.104
*Np	0	0	0.358	0.502	0.035	0.144
*Nk	2.747	0	1.450	0.317	2.090	0.151
*RT	4.000	0.989	2.961	0.915	2.948	0.685
*Rp	0.759	0.760	1.291	0.834	1.268	0.473
*Rk	3.205	0.229	1.669	0.081	1.680	0.424
<i>Unnatural markedness constraints</i>						
*[-voice,-son]	0					
*p	0.896					
*k	0					
*#[-voice,-son]	1.882					
*#p	1.053					
*#k	0.857					
DISAGREE(voice)/[-son]	4.955		4.698		4.484	0.688
DISAGREE(voice)/[+labial,-son]	0.345		0.475		0.875	0.468
DISAGREE(voice)/[+dorsal,-son]	2.624		1.019		1.145	0.488
CODACOND([-voice])/[-son]	0.015					
CODACOND([-voice])/[+labial,-son]	0					
CODACOND([-voice])/[+dorsal,-son]	2.033					
*VDV	0					
*VbV	0.609					
*VgV	0.008					
*ND	0.022		0.114	1.767		
*Nb	1.857		1.024	1.421		
*Ng	0.002		0.100	0.347		
*RD	1.858					
*Rb	0.002					
*Rg	1.854					

Table A11 The best performing models and the weights assigned to them by the learner.